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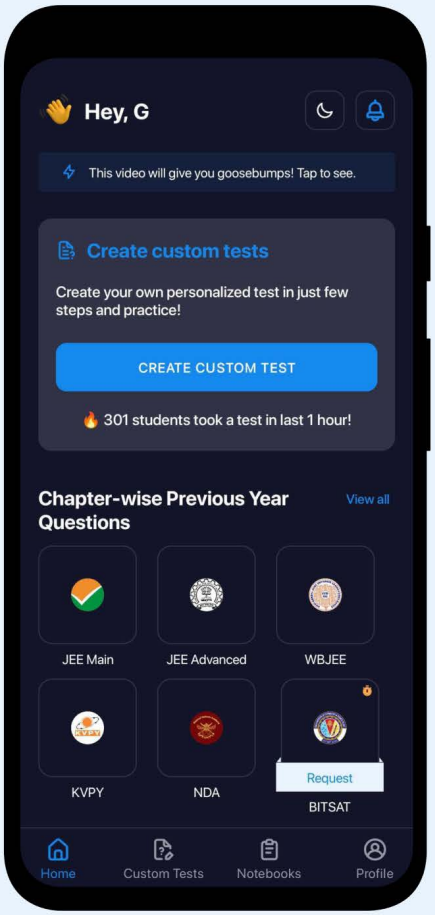

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TRIGONOMETRIC RATIOS

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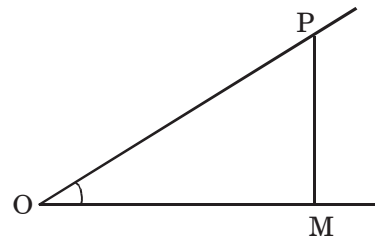
DEFINITION OF TRIGONOMETRIC RATIOS

Let a revolving line OP starts from OM and revolves into the position OP, thus tracing out the angle MOP.

Draw PM perpendicular to the initial line OM.

In the triangle MOP, OP is the hypotenuse, PM is the perpendicular, and OM is the base. The trigonomietrical ratios, or fuctions, of the angle MOP are defined as follows.



$\frac{MP}{OP}$, i.e., $\frac{Perp.}{Hyp.}$, is called the **Sine** of the angle AOP;

$\frac{OM}{OP}$, i.e., $\frac{Base}{Hyp.}$, is called the **Cosine** of the angle AOP;

$\frac{MP}{OM}$, i.e., $\frac{Perp.}{Base}$, is called the **Tangent** of the angle AOP;.

$\frac{OM}{MP}$, i.e., $\frac{Base}{Perp.}$, is called the **Cotangent** of the angle AOP;

$\frac{OP}{OM}$, i.e., $\frac{Hyp.}{Base}$, is called the **Secant** of the angle AOP;

$\frac{OP}{MP}$, i.e., $\frac{Hyp.}{Perp.}$, is called the **Cosecant** of the angle AOP;

Sign of Trigonometric Ratios

As we know that cartesian plane is divided into 4 quadrants, the sign of trigonometric ratios vary according to the quadrant

90

IInd quadrant
Only sinθ & cosecθ are positive

Ist quadrant
All T-ratios are positive (as both x & y are tue)

180 — O or 360

Only tan θ & cotθ are positive
IIIrd quadrant

Only cos θ & secθ are positive
IV quadrant

270

You can learn this thing by the following line

'After	**School**	**To**	**College"**
↓	↓	↓	↓
All	sin	tan	cos θ
T-ratios	T-ratio	T-ratios	T-ratios
	includes	includes	includes
	$\frac{1}{\sin\theta}$ = cosec θ	cot θ	sec θ

Note : The angle we talk about in trigonometry and all its formulas is in **Radians** and NOT in degrees. So we convert degree into radians whenever we are given question in degrees.

Relation between Degree & Radian

π radian = 180 degree

$\therefore$ 1 radian = $\frac{180}{\pi}$ degree

or 1 degree = $\frac{\pi}{180}$ radian

TRIGONOMETRIC RATIOS OF SOME ANGLES

Angle	0	30	45	60	90	120	135	150	180
Sine	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Cosine	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	–1
Tangent	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	–1	$-\frac{1}{\sqrt{3}}$	0
Cotangent	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$-\frac{1}{\sqrt{3}}$	–1	$-\sqrt{3}$	∞
Cosecant	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
Secant	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	–2	$-\sqrt{2}$	$-\frac{2}{\sqrt{3}}$	–1

TRIGONOMETRIC RATIOS FOR AN ANGLE OF ANY MAGNITUDE

Complementary Angles

Two angles are said to be complementary when their sum is equal to a right angle. Thus any angle θ and the angle $90 - \theta$ are complementary.

Supplementary angles

Two angles are said to be supplementary when their sum is equal to two right angles i.e., the supplement of any angle θ is $180 - \theta$.

Allied or Related Angles

The angles $\frac{1}{2}n\pi + \theta$ and $\frac{1}{2}n\pi - \theta$, where is any integer, are known as allied or related angles.

The trigonometric functions of these angles can be expressed as trigonometric functions of θ, with either a plus or a minus sign.The following working rules can be used in determining these functions.

for complimentary angles , T-ratio interchange

$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$, $\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$

$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$, $\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$

$\text{cosec}\left(\frac{\pi}{2} - \theta\right) = \sec\theta$, $\sec\left(\frac{\pi}{2} - \theta\right) = \text{cosec}\,\theta$

for supplementary angles, T-ratios remain same though value may become negative relative to quadrant

$\sin(\pi - \theta) = \sin\theta$ $\sin(\pi + \theta) = -\sin\theta$ {as in III quad sin is negative}

$\cos(\pi - \theta) = -\cos\theta$ $\cos(\pi + \theta) = -\cos\theta$

$\tan(\pi - \theta) = -\tan\theta$ $\tan(\pi + \theta) = +\tan\theta$

(apply the quadrant rule for the last two also)

TABLE OF FORMULAE FOR ALLIED ANGLES

	$180 - \theta$	$180 + \theta$	$360 - \theta$	$-\theta$	$90 - \theta$	$90 + \theta$	$270 - \theta$	$270 + \theta$
Sin	$\sin\theta$	$-\sin\theta$	$-\sin\theta$	$-\sin\theta$	$\cos\theta$	$\cos\theta$	$-\cos\theta$	$-\cos\theta$
Cos	$-\cos\theta$	$-\cos\theta$	$\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\sin\theta$	$\sin\theta$
Tan	$-\tan\theta$	$+\tan\theta$	$-\tan\theta$	$-\tan\theta$	$\cot\theta$	$-\cot\theta$	$\cot\theta$	$-\cot\theta$

For learning this table use the concept done earlier about quadrants

For Example sin (180 + θ) will enter III quadrant (assuming θ to be acute) and we know sin is negative in third quadrant

for ex. $\sin(180 + \theta) = -\sin\theta$

$\sin(270 + \theta) = -\cos\theta$

becuase (270 = 3 90) a complementary angle, so sinθ will change to cosθ, and 270 + θ lies in IV quadrant and sin is negative in IV quadrant, so –cosθ.

Note : We put the sign according to T-ratio on the left not on the R.H.S.

Also note T-ratios for – θ

$\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

$\tan(-\theta) = -\tan\theta$

BASIC FORMULAS

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$ or $\sec^2\theta - \tan^2\theta = 1$
3. $1 + \cot^2\theta = \text{cosec}^2\theta$ or $\text{cosec}^2\theta - \cot^2\theta = 1$

Some formulas derived from above reslts

- $\sin\theta = \pm\sqrt{1-\cos^2\theta}$ or $\cos\theta = \pm\sqrt{1-\sin^2\theta}$
- $\sec\theta + \tan\theta = \dfrac{1}{\sec\theta - \tan\theta}$
- $\text{cosec}\theta + \cot\theta = \dfrac{1}{\text{cosec}\,\theta - \cot\theta}$

Some alzebraic formulas useful in problems :

- $a^2 + b^2 = (a + b)^2 - 2ab$
- $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$
- $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$
- $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 0$

Componendo & dividendo :

If $\dfrac{p}{q} = \dfrac{a}{b}$, then by using componendo & dividendo

$$\frac{p+q}{p-q} = \frac{a+b}{a-b}$$

Illustration 1

Prove that $\sqrt{\dfrac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$

Solution :

$$\text{L.H.S.} = \sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sqrt{\frac{1+\sin\theta}{1-\sin\theta}\cdot\frac{1+\sin\theta}{1+\sin\theta}}$$

$$= \sqrt{\frac{(1+\sin\theta)^2}{1-\sin^2\theta}} = \sqrt{\frac{(1+\sin\theta)^2}{\cos^2\theta}} = \frac{1+\sin\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} = \sec\theta + \tan\theta = \text{R.H.S.}$$

Illustration 2

Show that (1 + cotA – cosec A) (1 + tanA + secA) = 2

Solution :

$$\text{L.H.S.} = (1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$$

$$= \left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right)\left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$$

$$= \left(\frac{\sin A + \cos A - 1}{\sin A}\right)\left(\frac{\cos A + \sin A + 1}{\cos A}\right)$$

$$= \frac{(\sin A + \cos A)^2 - 1^2}{\sin A \cos A}$$

$$= \frac{\sin^2 A + \cos^2 A + 2\sin A\cos A - 1}{\sin A \cos A}$$

$$= \frac{2\sin A\cos A}{\sin A\cos A} = 2 = \text{R.H.S.}$$

Illustration 3

Prove that $3(\sin x - \cos x)^4 + 4(\sin^6 x + \cos^6 x) + 6(\sin x + \cos x)^2 = 13$

Solution :

$$\text{L.H.S.} = 3[(\sin x - \cos x)^2]^2 + 4[(\sin^2 x)^3 + (\cos^2 x)^3)] + 6(\sin^2 x + \cos^2 x + 2\sin x \cos x)$$

$= 3(\sin^2x + \cos^2x - 2\sin x \cos x)^2 + 4(\sin^2x + \cos^2x)(\sin^4x + \cos^4x - \sin^2x \cos^2x)]$
$+ 6(\sin^2x + \cos^2x + 2\sin x \cos x)$

$= 3(1 - 2\sin x \cos x)^2 + 4(\sin^4x + \cos^4x) - \sin^2x \cos^2x) + 6(1 + 2\sin x.\cos x)$

$= 3(1 + 4\sin^2x \cos^2x - 4\sin x \cos x) + 4[(\sin^2x + \cos^2x)^2 - 2\sin^2x \cos^2x - \sin^2x \cos^2x]$
$+ 6 + 12\sin x \cos x$

$= 3 + 12\sin^2x \cos^2x - 12\sin x \cos x + 4(1 - 3\sin^2x \cos^2x) + 6 + 12\sin x \cos x$

$= 3 + 12\sin^2x \cos^2x + 4 - 12\sin^2x \cos^2x + 6 = 13$

Illustration 4

If $\tan^2\theta = 1 - e^2$, prove that $\sec\theta + \tan^3\theta \operatorname{cosec}\theta = (2 - e^2)^{\frac{3}{2}}$

Solution :

Given $\tan^2\theta = 1 - e^2$

Now, L.H.S. $= \sec\theta + \tan^3\theta \operatorname{cosec}\theta$

$= \sec\theta\left(1 + \tan^3\theta \dfrac{\operatorname{cosec}\theta}{\sec\theta}\right)$

$= \sec\theta\,(1 + \tan^3\theta.\cot\theta) = \sec\theta\,(1 + \tan^2\theta)$

$= \sec\theta \sec^2\theta$

$= \sec^3\theta = (\sec^2\theta)^{\frac{3}{2}} = (1 + \tan^2\theta)^{\frac{3}{2}}$

$= (1 + 1 - e^2)^{\frac{3}{2}} = (2 - e^2)^{\frac{3}{2}}$

Illustration 5

$\operatorname{Cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, eliminate θ.

Solution :

Given, $\operatorname{cosec}\theta - \sin\theta = m$ or, $\dfrac{1}{\sin\theta} - \sin\theta = m$

or, $\dfrac{1 - \sin^2\theta}{\sin\theta} = m$ or, $\dfrac{\cos^2\theta}{\sin\theta} = m$ ...(i)

Again $\sec\theta - \cos\theta = n$ or, $\dfrac{1}{\cos\theta} - \cos\theta = n$

or, $\dfrac{1 - \cos^2\theta}{\cos\theta} = n$ or, $\dfrac{\sin^2\theta}{\cos\theta} = n$ ...(ii)

From (i), $\sin\theta = \dfrac{\cos^2\theta}{m}$ ...(iii)

Putting in (ii), we get $\dfrac{\cos^4\theta}{m^2\cos\theta} = n$ or, $\cos^3\theta = m^2n$

$\therefore$ $\cos\theta = (m^2n)^{\frac{1}{3}}$ or, $\cos^2\theta = (m^2n)^{\frac{2}{3}}$ ...(iv)

From (iii), $\sin\theta = \dfrac{\cos^2\theta}{m} = \dfrac{(m^2n)^{\frac{2}{3}}}{m} = \dfrac{m^{\frac{4}{3}}n^{\frac{2}{3}}}{m} = m^{\frac{1}{3}}n^{\frac{2}{3}} = (mn^2)^{\frac{1}{3}}$

$\therefore$ $\sin^2\theta = (mn^2)^{\frac{2}{3}}$ ...(v)

adding (iv) and (v), we get

$$(m^2n)^{\frac{2}{3}} + (mn^2)^{\frac{2}{3}} = \cos^2\theta + \sin^2\theta$$

or, $(m^2n)^{\frac{2}{3}} + (mn^2)^{\frac{2}{3}} = 1.$

Problems based on limits between which $\sin\theta$, $\cos\theta$, $\sec\theta$, $\text{cosec}\theta$, lies or does not lie.

Working Rule : Use the following whichever is applicable.

(i) $-1 \le \sin\theta \le 1$ or , $|\sin\theta| \le 1$ or, $\sin^2\theta \le 1$

(ii) $-1 \le \cos\theta \le 1$ or, $\cos^2\theta \le 1$

(iii) $\sec\theta \le -1$ or, $\sec\theta \ge 1$ i.e. $\sec^2\theta \ge 1$

(iv) $\text{cosec}\theta \le 1$ or, $\text{cosec}\theta \ge 1$ i.e. $\text{cosec}^2\theta \ge 1$

Illustration 6

Is the equation $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ possible for real values of x and y ?

Solution :

Given, $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$

$\because$ $\sec^2\theta \ge 1$ $\therefore$ $\dfrac{4xy}{(x+y)^2} \ge 1$

or, $(x + y)^2 \le 4xy$ or, $(x + y)^2 - 4xy \le 0$ or, $(x - y)^2 \le 0$

But for real values of x and y, $(x - y)^2 < 0$

$\therefore$ $(x - y)^2 = 0$ $\therefore$ $x = y$ Also $x + y \ne 0 \Rightarrow x \ne 0.\ y \ne 0$

$\therefore$ given equation $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ is possible for real values of x and y only when x = y and $x \ne 0$

Illustration 7

Show that the equation $\sin\theta = x + \dfrac{1}{x}$ is impossible if x is real.

Solution :

Given $\sin\theta = x + \dfrac{1}{x}$

$\therefore$ $\sin\theta = x^2 + \dfrac{1}{x^2} + 2x.\dfrac{1}{x} = x^2 + \dfrac{1}{x^2} + 2 \ge 2$

which is not possible since $\sin^2\theta \le 1$

Illustration 8

If $\sin\theta + \sin^2\theta = 1$, then prove that $\cos^{12}\theta + 3\cos^{10}\theta + 3\cos^8\theta + \cos^6\theta - 1 = 0$

Solution :

Given $\sin\theta = 1 - \sin^2\theta = \cos^2\theta$. ...(1)

L.H.S. $= \text{Cos}^6\theta[\cos^2\theta + 1]^3 - 1$

$= \sin^3\theta\ (1 + \sin\theta)^3 - 1$ by (1)

$= (\sin\theta + \sin^2\theta)^3 - 1 = 1^3 - 1 = 0.$ by (1)

Illustration 9

If $t_n = \sin^n\theta + \cos^n\theta$, then $\dfrac{t_3 - t_5}{t_1} = \dfrac{t_5 - t_7}{t_3}$

Solution :

L.H.S. $= [(\sin^3\theta + \cos^3\theta) - (\sin^5\theta + \cos^4\theta) \div (\sin\theta + \cos\theta)$

$= [\sin^3\theta(1 - \sin^2\theta) + \cos^3\theta\ (1 - \cos^2\theta)] \div (\sin\theta + \cos\theta)$

$= \sin^2\theta\ \cos^2\theta\ [\sin\theta + \cos\theta) \div (\sin\theta + \cos\theta)$

$= \sin^2\theta\ \cos^2\theta.$

the L.H.S. is

$$\text{L.H.S.} = \frac{t_3 - t_5}{t_1}$$

now we will solve R.H.S.

$$\text{R.H.S.} = \frac{t_5 - t_7}{t_3}$$

$$= \frac{(\sin^5\theta + \cos^5\theta) - (\sin^7\theta + \cos^7\theta)}{(\sin^2\theta + \cos^3\theta)} = \frac{\sin^5\theta(1 - \sin^2\theta) + \cos^5\theta(1 - \cos^2\theta)}{\sin^3\theta + \cos^3\theta}$$

$$= \frac{\sin^5\theta\cos^2\theta + \cos^5\theta\sin^2\theta}{\sin^3\theta + \cos^3\theta} = \frac{\sin^2\theta\cos^2\theta(\sin^3\theta + \cos^3\theta)}{(\sin^3\theta + \cos^3\theta)}$$

= L.H.S.

Hence proved

Illustration 10

$$\left(\frac{1 + \sin\theta - \cos\theta}{1 + \sin\theta + \cos\theta}\right)^2 = \frac{1 - \cos\theta}{1 + \cos\theta}$$

Solution :

$$\text{L.H.S.} = \frac{(1 + \sin\theta)^2 + \cos^2\theta - 2\cos\theta(1 + \sin\theta)}{(1 + \sin\theta)^2 + \cos^2\theta + 2\cos\theta(1 + \sin\theta)}$$

$$= \frac{1 + \sin^2\theta + \cos^2\theta + 2\sin\theta - 2\cos\theta(1 + \sin\theta)}{1 + \sin^2\theta + \cos^2\theta + 2\sin\theta + 2\cos\theta\,(1 + \sin\theta)}$$

$$= \frac{2(1 + \sin\theta) - 2\cos\theta(1 + \sin\theta)}{2(1 + \sin\theta) + 2\cos\theta(1 + \sin\theta)} = \frac{1 - \cos\theta}{1 + \cos\theta}$$

We have cancelled $2(1 + \sin\theta)$ from N^r and D^r.

Illustration 11

$$\left(\frac{1}{\sec^2\alpha - \cos^2\alpha} + \frac{1}{\operatorname{cosec}^2\alpha - \sin^2\alpha}\right) \times \sin^2\alpha\cos^2\alpha = \frac{1 - \sin^2\alpha\cos^2\alpha}{2 + \sin^2\alpha\cos^2\alpha}$$

Solution :

$$\frac{1}{\sec^2\alpha - \cos^2\alpha} = \frac{\cos^2\alpha}{1 - \cos^4\alpha}$$

$$= \frac{\cos^2\alpha}{(1-\cos^2\alpha)(1+\cos^2\alpha)}$$

$$= \frac{\cos^2\alpha}{\sin^2\alpha(1+\cos^2\alpha)}$$

Similarly

$$\frac{1}{\text{cosec}^2\alpha - \sin^2\alpha} = \frac{\sin^2\alpha}{\cos^2\alpha(1+\sin^2\alpha)}$$

Hence,

$$\text{L.H.S.} = \left[\frac{\cos^2\alpha}{\sin^2\alpha(1+\cos^2\alpha)} + \frac{\sin^2\alpha}{\cos^2\alpha(1+\sin^2\alpha}\right]\sin^2\alpha\cos^2\alpha$$

$$= \frac{\cos^4\alpha}{1+\cos^2\alpha} + \frac{\sin^4\alpha}{1+\sin^2\alpha}$$

$$= [\sin^2\alpha\cos^2\alpha(\sin^2\alpha + \cos^2\alpha) + (\cos^4\alpha + \sin^4\alpha)] \div (1 + \cos^2\alpha + \sin^2\alpha + \cos^2\alpha\sin^2\alpha)$$

$$= \frac{\sin^2\alpha\cos^2\alpha + (\cos^2\alpha + \sin^2\alpha)^2 - 2\sin^2\alpha\cos^2\alpha}{2+\cos^2\alpha\sin^2\alpha}$$

$$= \frac{1-\sin^2\alpha\cos^2\alpha}{2+\cos^2\alpha\sin^2\alpha} = \text{R.H.S.}$$

Illustration 12

If $a\cos\theta - b\sin\theta = c$, **show that** $a\sin\theta + b\cos\theta = \pm\sqrt{a^2 + b^2 - c^2}$

Solution :

Square $a\cos\theta - b\sin\theta = c$. Then

$\therefore \quad a^2\cos^2\theta + b^2\sin^2\theta - 2ab\sin\theta\cos\theta = c^2$

or $\quad a^2(1-\sin^2\theta) + b^2(1-\cos^2\theta) - 2ab\sin\theta\cos\theta = c^2$

$\therefore \quad a^2 + b^2 - c^2 = a^2\sin^2\theta + b^2\cos^2\theta + 2ab\sin\theta\cos\theta$

or $\quad (a\sin\theta + b\cos\theta)^2 = a^2 + b^2 - c^2$

$\therefore \quad a\sin\theta + b\cos\theta = \pm\sqrt{(a^2 + b^2 - c^2)}$

Illustration 13

$\frac{ax}{\cos\theta} + \frac{by}{\sin\theta} = (a^2 - b^2)$, and $\frac{ax\sin\theta}{\cos^2\theta} - \frac{by\cos\theta}{\sin^2\theta} = 0$, **show that** $(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}$

Solution :

From the second relation, we get

$$\frac{\sin^2\theta\sin\theta}{\cos^2\theta\cos\theta} = \frac{by}{ax}$$

$\tan^3\theta = by/ax \quad \therefore \quad \tan\theta = (by)^{1/3}/(ax)^{1/3}$

$$\therefore \quad \sin\theta = \frac{(by)^{1/3}}{[(ax)^{2/3} + (by)^{2/3}]^{1/2}},$$

$$\cos\theta = \frac{(ax)^{1/3}}{[(ax)^{2/3} + (by)^{2/3}]^{1/2}}$$

Putting for $\sin\theta$ and $\cos\theta$ in

$\frac{ax}{\cos\theta} + \frac{by}{\sin\theta} = a^2 - b^2$, we get

$[(ax)^{2/3} + (by)^{2/3}]^{1/2}\ [ax/(ax)^{1/3} + by/(by)^{1/3}] = a^2 - b^2$

or $[(ax)^{2/3} + (by)^{2/3}]^{1/2}\ [(ax)^{2/3} + (by)^{2/3}] = a^2 - b^2$

or $[(ax)^{2/3} + (by)^{2/3}]^{3/2}. = a^2 - b^2$

or $(ax)^{2/3} _ (by)^{2/3} = (a^2 - b^2)^{2/3}$

Illustration 14

If $m^2 + m'^2 + 2mm'\cos\theta = 1$, $n^2 + n'^2 + 2nn'\cos\theta = 1$, **and** $mn + m'n' + (mn' + m'n)\cos\theta = 0$, **prove that** $m^2 + n^2 = \text{cosec}^2\theta$.

Solution :

The given relation can be written as

$(m' + m\cos\theta)^2 = 1\ m^2 - m^2\cos^2\theta = 1$

or $(m' + m\cos\theta)^2 = 1 - m^2\sin^2\theta$ ...(1)

Similarly $(n' + n\cos\theta)^2 = 1 - n^2\sin^2\theta$ ...(2)

Now $(m' + m\cos\theta)(n' + n\cos\theta)$

$= m'n' + (mn' + m'n)\cos\theta + mn\cos^2\theta$

$= -mn + mn\cos^2\theta$ by given relation

$= mn(1 - \cos^2\theta) = -mn\sin^2\theta$

Now squaring both sides, we get

or $(m' + m\cos\theta)^2 (n' + n\cos\theta)^2 = m^2n^2 \sin^4\theta$ ...(3)

Hence substituting from (1) and (2) in (3), we get

$(1 - m^2 \sin^2\theta)(1 - n^2 \sin^2\theta) = m^2n^2 \sin^4\theta$

or $(m^2 + n^2) \sin^2\theta = 1$ i.e. $m^2 + n^2 = \text{cosec}^2\theta$.

(b) Eliminate θ from the equations

$a \cos\theta + b \sin\theta = c$ and

$a \cos^2\theta + 2a \sin\theta \cos\theta + b \sin^2\theta = c$

Illustration 15

Prove that tan 1 tan 2 tan 3 ... tan 89 =1

Solution :

We will use the property of complementary angles to solve this problem.

We know $\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$

So now tan 89 = tan(90 – 1) = cot 1

tan 88 = tan(90 – 2) = cot 2

So going in this manner

(tan 1 cot 1) (tan 2 cot 2) ... (tan 44 cot 44 tan 45)

= tan 45 = 1

as only tan45 term will be left

Illustration 16

Prove that $\sin^2 5 + \sin^2 10 + \sin^2 15 + \ldots \sin^2 85 + \sin^2 90 = 9\frac{1}{2}$

Solution :

This question will also be solved on the same lines as that of previous questions

sin 5 = sin(90 – 45) = cos 85

Similarly sin 10 = sin(90 – 80) = cos 80

sin 40 = sin(90 – 50) = cos 50

& the middle term is sin 45

$(\sin^2 85 + \cos^2 85) + (\sin^2 80 + \cos^2 80) \ldots (\sin^2 50 + \cos^2 50) + \sin^2 45$

from 85 to 50 with a difference of 5 there are 8 terms (pairwise) & we know $\sin^2\theta + \cos^2\theta = 1$

Hence

$= 8 + \sin^2 45 = 8 + \frac{1}{2} = \frac{17}{2}$

Sum and difference formulas

- $\sin(A + B) = \sin A \cos B + \sin B \cos A$
- $\sin(A - B) = \sin A \cos B - \sin B \cos A$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Note the cos formulas, they have (–) sign for sum formula and (+) ve sign for difference.

- $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

- $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

where $A, B \neq \left(n + \dfrac{1}{2}\right)\pi$

- $\cot(A + B) = \dfrac{\cot A \cot B - 1}{\cot A + \cot B}$

- $\cot(A - B) = \dfrac{\cot A \cot B + 1}{\cot B - \cot A}$

where $A, B \neq n\pi$

Cot formula's are opposite of tan formulas, the denominator & numerator part are switched.

Some Important deductions

- $\sin(A+B)\sin(A - B) = \sin^2 A - \sin^2 B$
  $= \cos^2 B - \cos^2 A$
- $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$
  $= \cos^2 B - \sin^2 A$

- $\tan A \pm \tan B = \dfrac{\sin(A \pm B)}{\cos A \cos B}$

- $\cot A \pm \cot B = \dfrac{\sin(B \pm A)}{\sin A \sin B}$

- $\tan\left(\dfrac{\pi}{4} + \theta\right) = \dfrac{1 + \tan\theta}{1 - \tan\theta}$ & $\tan\left(\dfrac{\pi}{4} - \theta\right) = \dfrac{1 - \tan\theta}{1 + \tan\theta}$

This formula will be used in various number of questions.

Trignometric Ratio of multiple angles

- $\sin 2A = 2 \sin A \cos A$
- $\cos 2A = \cos^2 A - \sin^2 A$
  $= 2\cos^2 A - 1$
  $= 1 - 2\sin^2 A$
- $\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$
- $\sin 3A = 2 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3\cos A$
- $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Some formulas related to half angles

- $\dfrac{1 - \cos\theta}{\sin\theta} = \tan\dfrac{\theta}{2}$
- $\dfrac{1 + \cos\theta}{\sin\theta} = \cot\dfrac{\theta}{2}$
- $\dfrac{1 - \cos\theta}{1 + \cos\theta} = \tan^2\dfrac{\theta}{2}$
- $\dfrac{1 + \cos\theta}{1 - \cos} = \cot^2\dfrac{\theta}{2}$

Other deductions

Sometimes even after remembering all these above formulas we are not able to recognise them in questions. So look into these formulas also

Conversion of squares & cubes into multiple angles :

- $\sin^2 A = \dfrac{1}{2}(1 - \cos 2A)$
- $\cos^2 A = \dfrac{1}{2}(1 + \cos 2A)$
- $\sin^3 A = \dfrac{1}{4}(3\sin A - \sin 3A)$
- $\cos^3 A = \dfrac{1}{4}(\cos 3A + 3\cos A)$

Other formulas

$$\sin A = \frac{2\tan A/2}{1+\tan^2 A/2}$$

$$\cos A = \frac{1-\tan^2 A/2}{1+\tan^2 A/2}$$

TRANSFORMATION FORMULAS

1. Transformation of product into sum or differences
   - $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
   - $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
   - $2 \cos A \cos B = \cos(A + B) + \cos (A - B)$
   - $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
2. Transformation of sum & differences into product
   - $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$
   - $\sin A - \sin B = 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$
   - $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$
   - $\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$

Illustration 18

Prove that $\frac{\sin(B-C)}{\cos B \cos C} + \frac{\sin(C-A)}{\cos C \cos A} + \frac{\sin(A-B)}{\cos A \cos B} = 0$

Solution : First term of L.H.S.

$$= \frac{\sin(B-C)}{\cos B \cos C} = \frac{\sin B \cos C - \cos B \sin C}{\cos B \cos C}$$

$$= \frac{\sin B \cos C}{\cos B \cos C} - \frac{\cos B \sin C}{\cos B \cos C} = \tan B - \tan C$$

Similarly, second term of L.H.S. = tan C – tan A

and, 3rd term of L.H.S. = tan A – tan B

Now, L.H.S. = (tanB – tanC) + (tanC – tanA) + (tanA – tanB) = 0

Illustration 19

Show that $\cot\left(\frac{\pi}{4}+x\right).\cot\left(\frac{\pi}{4}-x\right)=1$

Solution : L.H.S. $= \dfrac{\cos\left(\frac{\pi}{4}+x\right)\cos\left(\frac{\pi}{4}-x\right)}{\sin\left(\frac{\pi}{4}+x\right)\sin\left(\frac{\pi}{4}-x\right)}$

$$= \frac{\cos^2\frac{\pi}{4}-\sin^2 x}{\sin^2\frac{\pi}{4}-\sin^2 x} = \frac{\frac{1}{2}-\sin^2 x}{\frac{1}{2}-\sin^2 x} = 1$$

Illustration 20

If $\sin\alpha\ \sin\beta - \cos\alpha\ \cos\beta + 1 = 0$, prove that $1 + \cot\alpha\ \tan\beta = 0$

Solution :

Given, $\sin\alpha\ \sin\beta - \cos\alpha\ \cos\beta + 1 = 0$

or, $\cos\alpha\ \cos\beta - \sin\alpha\ \sin\beta = 1$

or, $\cos(\alpha + \beta) = 1$

Now, $1 + \cot\alpha\ \tan\beta = 1 + \dfrac{\cos\alpha}{\sin\alpha}.\dfrac{\sin\beta}{\cos\beta}$

$$= \frac{\sin\alpha\cos\beta+\cos\alpha\sin\beta}{\sin\alpha\cos\beta} = \frac{\sin(\alpha+\beta)}{\sin\alpha\cos\beta}$$

$$= \frac{0}{\sin\alpha\cos\beta} = 0 \quad [\therefore\ \sin^2(\alpha+\beta) = 1-\cos^2(\alpha+\beta) = 1-1 = 0]$$

Illustration 21

If $\tan\beta = \dfrac{n\sin\alpha\cos\alpha}{1-n\sin^2\alpha}$, prove that $\tan(\alpha-\beta) = (1-n)\tan\alpha$

Solution : $\tan\beta = \dfrac{n\sin\alpha\cos\alpha}{1-n\sin^2\alpha}$

$$= \frac{\frac{n\sin\alpha\cos\alpha}{\cos^2\alpha}}{\frac{1}{\cos^2\alpha}-\frac{n\sin^2\alpha}{\cos^2\alpha}} \quad \text{[dividing numerator \& denominator by } \cos^2\alpha]$$

$$= \frac{n\tan\alpha}{\sec^2\alpha - n\tan^2\alpha} = \frac{n\tan\alpha}{1+\tan^2\alpha - n\tan^2\alpha}$$

$$= \frac{n\tan\alpha}{1+(1-n)\tan^2\alpha} \qquad ...(i)$$

Now, L.H.S. $= \tan(\alpha - \beta) = \dfrac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta}$

$$= \frac{\tan\alpha - \dfrac{n\tan\alpha}{1+(1-n)\tan^2\alpha}}{1+\tan\alpha\dfrac{n\tan\alpha}{1+(1-n)\tan^2\alpha}} \qquad \text{[From (i)]}$$

$$= \frac{\tan\alpha + (1-n)\tan^3\alpha - n\tan\alpha}{1+(1-n)\tan^2\alpha + n\tan^2\alpha} = \frac{(1-n)\tan\alpha + (1-n)\tan^3\alpha}{1+\tan^2\alpha}$$

$$= \frac{(1-n)\tan\alpha(1+\tan^2\alpha)}{1+\tan^2\alpha} = (1-n)\tan\alpha$$

Illustration 22

If $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = -\dfrac{3}{2}$, prove that $\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 0$

Solution :

Given, $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = -\dfrac{3}{2}$

or, $3 + 2\cos(\beta-\gamma) + 2\cos(\gamma-\alpha) + 2\cos(\alpha-\beta) = 0$

or, $3 + 2(\cos\beta\cos\gamma + \sin\beta\sin\gamma) + 2(\cos\gamma\cos\alpha + \sin\gamma\sin\alpha) + 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = 0$

or, $(\cos^2\alpha + \sin^2\alpha) + (\cos^2\beta + \sin^2\beta) + (\cos^2\gamma + \sin^2\gamma) + 2(\cos\beta\cos\gamma + \sin\beta\sin\gamma)$
$+ 2(\cos\gamma\cos\alpha + \sin\gamma\sin\alpha) + 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = 0$

or, $(\cos^2\alpha + \cos^2\beta + \cos^2\gamma + 2\cos\alpha\cos\beta + 2\cos\beta\cos\gamma + 2\cos\gamma\cos\gamma) + (\sin^2\alpha + \sin^2\beta + \sin^2\gamma$
$+ 2\sin\alpha\sin\beta + 2\sin\beta\sin\gamma + 2\sin\gamma\sin\alpha) = 0$

or, $(\cos\alpha + \cos\beta + \cos\gamma)^2 + (\sin\alpha + \sin\beta + \sin\gamma)^2 = 0$

which is possible only when

$\cos\alpha + \cos\beta + \cos\gamma = 0$ and $\sin\alpha + \sin\beta + \sin\gamma = 0$

Illustration 23

If cos (A + B) sin (C + D) = cos (A – B) sin (C – D) prove that cotA cotB cotC = cotD.

Solution :

We have cos (A + B) sin (C + D) = cos(A – B) sin (C – D)

i.e. $\dfrac{\cos(A-B)}{\cos(A+B)} = \dfrac{\sin(C+D)}{\sin(C-D)}$

$\Rightarrow \dfrac{\cos(A-B)+\cos(A+B)}{\cos(A-B)-\cos(A+B)} = \dfrac{\sin(C+D)+\sin(C-D)}{\sin(C+D)-\sin(C-D)}$

or, $\dfrac{2\cos A\cos B}{2\sin A\sin B} = \dfrac{2\sin C\cos D}{2\cos C\sin D}$

$\Rightarrow$ cotA cotB = tanC cotD or Cot A cot B cot C = cot D

Illustration 24

Prove that tan 70 – tan 20 = 2 tan 50

Solution :

$\therefore$ 70 = 50 + 20

$\therefore$ $\tan 70 = \dfrac{\tan 50° + \tan 20°}{1 - \tan 50° \tan 20°}$

or tan 70 – tan 70 tan 50 tan 20 = tan 50 + tan 20

or tan 70 – tan 20 = tan 50 + tan 50 = 2 tan 50

[$\therefore$ tan 70 tan 20 = tan (90 – 20) tan 20 = cot 20 tan 20 = 1]

Alt. $\tan 70 - \tan 20 = \dfrac{\sin(70° - 20°)}{\cos 70° \cos 20°}$

$= \dfrac{2\sin 50°}{2\sin 20° \cos 20°}$ **{Comp. Rule)**

$= \dfrac{2\sin 50°}{\sin 40°} = \dfrac{2\sin 50°}{\cos 50°} = 2\tan 50°$

Illustration 25

tan 3A tan 2A tan A = tan 3A – tan 2A – tan A

Solution :

Rearranging the given relation, we have to prove that

tan A + tan 2A = tan 3A (1 – tan A tan 2A)

or $\dfrac{\tan A + \tan 2A}{1 - \tan A \tan 2A} = \tan 3A$

or $\tan (A + 2A) = \tan 3A$, which is true.

Illustration 26

Prove that $\tan (\pi/4 + \theta) - \tan (\pi/4 - \theta) = 2 \tan 2\theta$)

Solution :

$$\tan\left(\frac{\pi}{4} + \theta\right) = \frac{\tan(\pi/4) + \tan\theta}{1 - \tan(\pi/4) + \tan\theta} = \frac{1 + \tan\theta}{1 - \tan\theta}$$

Similarly, $\tan\left(\dfrac{\pi}{4} - \theta\right) = \dfrac{1 - \tan\theta}{1 + \tan\theta}$

$$\therefore \quad \text{L.H.S.} = \frac{1 + \tan\theta}{1 - \tan\theta} - \frac{1 - \tan\theta}{1 + \tan\theta} = \frac{(1 + \tan\theta)^2 - (1 - \tan\theta)^2}{1 - \tan^2\theta}$$

$$= \frac{4\tan\theta}{1 - \tan^2\theta} = 2.\frac{2\tan\theta}{1 - \tan^2\theta} = 2\tan 2\theta$$

Illustration 27

$$\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \sqrt{\left(\frac{1 + \sin\theta}{1 - \sin\theta}\right)} = \tan\theta + \sec\theta$$

Solution : L.H.S. $= \dfrac{1 + \tan(\theta/2)}{1 - \tan(\theta/2)} = \dfrac{\cos(\theta/2) + \sin(\theta/2)}{\cos(\theta/2) - \sin(\theta/2)}$

$$= \left[\frac{\{\cos(\theta/2) + \sin(\theta/2)\}^2}{\{\cos(\theta/2) - \sin(\theta/2)\}^2}\right]^{1/2} = \left[\frac{\cos^2(\theta/2) + \sin^2(\theta/2) + 2\sin(\theta/2)\cos(\theta/2)}{\cos^2(\theta/2) + \sin^2(\theta/2) - 2\sin(\theta/2)\cos(\theta/2)}\right]^{1/2}$$

$$= \sqrt{\left(\frac{1 + \sin\theta}{1 - \sin\theta}\right)} = \frac{1 + \sin\theta}{\sqrt{[(1 + \sin\theta)(1 - \sin\theta)]}}$$

$$= \frac{1 + \sin\theta}{\sqrt{(1 - \sin^2\theta)}} = \frac{1 + \sin\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} = \sec\theta + \tan\theta$$

Illustration 28

tan A + 2 tan 2A + 4 tan 4A + 8 cot 8A = cot A

Solution :

tan A + 2 tan 2A + 4 tan 4A + 8 cot 8A = cot A

shift tan A to R.H.S.

now R.H.S.

$= \cot A - \tan A$

$$= \frac{\cos A}{\sin A} - \frac{\sin A}{\cos A} = \frac{\cos^2 A - \sin^2 A}{\sin A \cos A}$$

$$= \frac{2\cos 2A}{\sin 2A} = 2 \cot 2A$$

Tip : Remember this formula

$\boxed{\cot A - \tan A = 2 \cot 2A}$

now question becomes

2 tan 2A + 4 tan 4A + 8 cot 8A = 2 cot 2A

$\Rightarrow$ 4 tan 4A + 8 cot 8A = 2 (cot 2A – tan 2A)

using the same formula

$\Rightarrow$ 4 tan 4A + 8 cot 8A = 4 cot 4A

$\Rightarrow$ 8 cot 8A = 4 (cot 4A – tan 4A)

now R.H.S.

= 4 (cot 4A – tan 4A)

= 8 cot 8A

= R.H.S.

Illustration 29

Prove sin A sin (60 – A) sin (60 + A) = $\frac{1}{4}$ sin 3A

Solution :

$$\sin A \sin (60 - A) \sin (60 + A) = \frac{1}{4} \sin 3A$$

L.H.S. = sin A sin (60 – A) sin (60 + A)

here using the formula

$\sin (A + B) \sin (A - B) = \sin^2 A - \sin^2 B$

$= \sin A (\sin^2 60 - \sin^2 A)$

$$= \sin A\left(\frac{3}{4} - \sin^2 A\right) = \frac{3\sin A - 4\sin^3 A}{4}$$

but now $3\sin A - 4\sin^3 A = \sin 3A$

$$\therefore \text{ L.H.S.} = \frac{\sin 3A}{4}$$

$$= \text{R.H.S.}$$

Hence proved.

Illustration 30

$$\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$$

Solution :

$7 + 1 = 3 + 5$

$$\text{L.H.S.} = \frac{(\sin A + \sin 7A) + (\sin 3A + \sin 5A)}{(\cos A + \cos 7A) + (\cos 3A + \cos 5A)}$$

$$= \frac{2\sin 4A \cos 3A + 2\sin 4A \cos A}{2\cos 4A \cos 3A + 2\cos 4A \cos A}$$

$$= \frac{\sin 4A\,(\cos 3A + \cos A)}{\cos 4A(\cos 3A + \cos A)} = \tan 4A$$

Illustration 31

$$\frac{\cos 3A + 2\cos 5A + \cos 7A}{\cos A + 2\cos 3A + \cos 5A} = \frac{\cos 5A}{\cos 3A} = \cos 2A - \sin 2A \tan 3A.$$

Solution :

$$\text{L.H.S.} = \frac{(\cos 3A + \cos 7A) + 2\cos 5A}{(\cos A + \cos 5A + 2\cos 3A}$$

$$= \frac{2\cos 5A \cos 2A + 2\cos 5A}{2\cos 3A \cos 2A + 2\cos 3A} = \frac{\cos 5A}{\cos 3A}$$

$$= \frac{\cos(3A + 2A)}{\cos 3A} = \frac{\cos 3A \cos 2A}{\cos 3A} - \frac{\sin 3A \sin 2A}{\cos 3A}$$

$$= \cos 2A - \tan 3A \sin 2A.$$

Illustration 32

$$\frac{\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10}{\cos 5\theta + 5\cos 3\theta + 10\cos\theta} = 2\cos\theta$$

Solution :

In the N^r write 6 as 1 + 5 and 15 as 10 + 5

$N^r = \cos 6\theta + \cos 4\theta + 5\cos 4\theta + 5\cos 2\theta + 10\cos 2\theta + 10$

$= 2\cos 5\theta \cos\theta + 5.2\cos 3\theta\cos\theta + 10.2\cos^2\theta$

$= 2\cos\theta\,[\cos 5\theta + 5\cos 3\theta + 10\cos\theta]$

$= 2\cos\theta\,[D^r]$

$$\therefore \quad \frac{N^r}{D^r} = 2\cos\theta$$

Alt. Bring 2 cos θ from R.H.S. with D^r in L.H.S. and using

$2\cos A\cos B = \cos(A + B) + \cos(A - B)$,

we get $D^r = (\cos 6\theta + \cos 4\theta) + 5(\cos 4\theta + \cos 2\theta) + 10(1 + \cos 2\theta)$

$= \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10 = N^r$

$\therefore \quad N^r / D^r = 1$

Illustration 33

$\cos\alpha + \cos\beta + \cos\gamma + \cos(\alpha + \beta + \gamma) = 4\cos\{(\alpha + \beta)/2\}\cos\{(\beta + \gamma)/2\}\cos\{(\gamma + \alpha)/2\}$

Solution :

$$\text{L.H.S.} = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} + 2\cos\frac{\alpha+\beta+2\gamma}{2}\cos\frac{\alpha+\beta}{2}$$

$$= 2\cos\frac{\alpha+\beta}{2}\left[\cos\frac{\alpha-\beta}{2} + \cos\frac{\alpha+\beta+2\gamma}{2}\right]$$

$$= 2\cos\frac{\alpha+\beta}{2}.2\cos\frac{\alpha+\gamma}{2}\cos\left(-\frac{\gamma+\beta}{2}\right) = 4\cos\frac{\alpha+\beta}{2}\cos\frac{\beta+\gamma}{2}\cos\frac{\gamma+\alpha}{2} \quad \{\because \cos(-\theta) = \cos\theta\}$$

Illustration 34

$$\left(\frac{\cos A + \cos B}{\sin A - \sin B}\right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B}\right)^n = 2\cot^n\left(\frac{A-B}{2}\right)$$ **or 0 according as n is even or odd integer.**

Solution :

$$\left[\frac{2\cos\{(A+B)/2\}\cos\{A-B)/2\}}{2\sin\{(A-B)/2\}\cos\{A+B)/2\}}\right]^n + \left[\frac{2\sin\{(A+B)/2\}\cos\{(A-B)/2\}}{2\sin\{(A+B)/2\}\sin\{(B-A)/2\}}\right]^n$$

$= \cot^n \{(A - B)/2\} + [- \cot \{(A - B)/2\}]n$

$= 2 \cot^n \{(A - B)/2$ or 0

according as n is even or odd as

$(-1)^n = 1$ or -1

Illustration 35

$$\frac{\cos 2A \cos 3A - \cos 2A \cos 7A + \cos A \cos 10A}{\sin 4A \sin 3A - \sin 2A \sin 5A + \sin 4A \sin 7A} = \cot 6A \cot 5A$$

Solution :

Multiply above and below by 2 and apply $2\cos A \cos B = \cos (A + B) + \cos (A - B)$ and $2 \sin A \sin B = \cos (A - B) - \cos (A + B)$

$$\therefore \quad \text{L.H.S.} = \frac{(\cos 5A + \cos A) - (\cos 9A + \cos 5A) + (\cos 11A + \cos 9A)}{(\cos A - \cos 7A) - (\cos 3A - \cos 7A) + (\cos 3A - \cos 11A)}$$

$$= \frac{\cos A + \cos 11A}{\cos A - \cos 11A} = \frac{2\cos 6A \cos 5A}{2\sin 6A \sin 5A}$$

$= \cot 6A \cot 5A.$

Illustration 36

If $2 \cos A = x + 1/x$, $2 \cos B = y + 1/y$, show that $2 \cos (A - B) = x/y + y/x$.

Solution :

$$2 \cos A = x + \frac{1}{x} \qquad \therefore \qquad x^2 - 2x \cos A + 1 = 0$$

$\therefore \quad x = \cos A + i \sin A = e^{iA}, \; y = e^{iB}$

$$\frac{x}{y} + \frac{y}{x} = \frac{e^{iB}}{e^{iA}} + \frac{e^{iA}}{e^{iB}} = e^{i(A - B)} + e^{-i}(A - B)$$

$= 2 \cos (A - B)$

$\therefore \quad e^{i\theta} = \cos \theta + i \sin \theta, \; e^{-i\theta} = \cos \theta - i \sin \theta$

Illustration 37

If α and β are the solutions of $a \cos \theta + b \sin \theta = c$, then show that

$\sin \alpha + \sin \beta = 2bc/(a^2 + b^2)$, $\sin \alpha \sin \beta = (c^2 - a^2)/(a^2 + b^2)$

Solution :

From the given relation we have

$a \cos \theta = c - b \sin \theta$. Square and change in terms of $\sin \theta$

$a^2 (1 - \sin^2 \theta) = c^2 - 2bc \sin \theta + b^2 \sin^2 \theta$

$\therefore \quad (a^2 + b^2) \sin^2 \theta - 2bc \sin \theta + (c^2 - a^2) = 0$

Its roots are $\sin \alpha$ and $\sin \beta$ as α and β are the values of θ as given.

$$\therefore \quad \sin \alpha + \sin \beta = \text{sum of roots} = \frac{2bc}{a^2 + b^2}$$

$$\sin \alpha \sin \beta = \text{product of roots} = \frac{c^2 - a^2}{a^2 + b^2}$$

Illustration 38

If $a \cos 2\theta + b \sin 2\theta = c$ has α and β as its solutions, then prove that

$\tan \alpha + \tan \beta = 2b/(c + a)$, $\tan \alpha \tan \beta = (c - a)/(c + a)$

Solution :

Given $a \cos 2\theta + b \sin 2\theta = c$

putting $\cos 2\theta = \dfrac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ & $\sin 2\theta = \dfrac{2 \tan \theta}{1 + \tan^2 \theta}$

now given equation becomes

$$a\left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}\right) + b\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right) = c$$

$a (1 - \tan^2\theta) + 2b \tan \theta = c (1 + \tan^2 \theta)$

$\Rightarrow \quad \tan^2\theta (c + a) - 2b \tan \theta + (c - a) = 0$

now this is quadratic in $\tan \theta$ & roots are given to be α & β

Hence $\quad \tan \alpha + \tan \beta = \dfrac{2b}{c + a}$

& $\quad \tan \alpha \tan \beta = \dfrac{c - a}{c + a}$

Illustration 39

If $\theta_1, \theta_2, \theta_3, \theta_4$ be the four values of q which satisfy the equation $3 \tan 3\theta = \tan (45 + \theta)$, then prove that $\Sigma \tan \theta_1 = 0$

Solution :

equation given is $3 \tan 3\theta = \tan (45 + \theta)$

putting $\tan 3\theta = \dfrac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$

& $\tan(45 + \theta) = \dfrac{1+\tan\theta}{1-\tan\theta}$

& $\tan\theta = t$

now equation becomes

$$\Rightarrow \qquad 3\left(\frac{3t - t^3}{1 - 3t^2}\right) = \frac{1+t}{1-t}$$

reducing this, equation becomes

$$3t^4 - 6t^2 + 8t - 1 = 0$$

Now $\theta_1\ \theta_2\ \theta_3\ \theta_4$ are solution to the equation

$$3\tan^4\theta - 6\tan^2\theta + 8\tan\theta - 1 = 0$$

$$\text{sum of roots} = \frac{-\text{coeff. of } \tan^3}{\text{coeff. of } \tan^4}$$

$= 0$, as there is no $\tan^3$ term

$\therefore\ \tan\theta_1 + \tan\theta_2 + \tan\theta_3 + \tan\theta_3 = 0$

Illustration 40

Find the max. and min. value of $7\cos\theta + 24\sin\theta$

Solution :

In $7\cos\theta + 24\sin\theta$,

take $7 = r\cos\alpha$ and $24 = r\sin\alpha$.

$\therefore\quad r = \sqrt{(7^2 + 24^2)} = 25$

$\therefore\quad$ L.H.S. $= r[\cos\theta\cos\alpha + \sin\theta\sin\alpha]$

$= 25\cos(\theta - \alpha)$

Now max. and min. values of $\cos(\theta - \alpha)$ are 1 and – 1.

Therefore the max. and min. values of given expression are 25 and – 25 respectively.

Illustration 41

Prove that $5\cos\theta + 3\cos(\theta + \pi/3) + 3$ lies between – 4 and 10.

Solution :

The given expression is

$$5\cos\theta + 3(\cos\theta\cos 60 - \sin\theta\sin 60) + 3 = \frac{1}{2}[(13\cos\theta - 3\sqrt{3}\sin\theta)] + 3$$

Put $13 = r\cos\alpha$, $3\sqrt{3} = r\sin\alpha$.

$\therefore \quad r = \sqrt{169 + 27} = \sqrt{196} = 14$

$\therefore$ Given expression $= \frac{r}{2}[\cos(\theta + \alpha)] + 3 = 7\cos(\theta + \alpha) + 3$

Hence max. and min. values of expression are 7 + 3 and – 7 + 3 i.e. 10 and – 4 respectively.

Illustration 42

If $\tan\frac{\theta}{2} = \sqrt{\left(\frac{1-e}{1+e}\right)}\tan\frac{\phi}{2}$, prove that $\cos\phi = \frac{\cos\theta - e}{1 - e\cos\theta}$

Solution :

We know that

$$\cos A = \frac{1 - \tan^2(A/2)}{1 + \tan^2(A/2)}$$

$$\therefore \quad \cos\phi = \frac{1 - \tan^2(\phi/2)}{1 + \tan^2(\phi/2)}$$

$$= \frac{1 - \{(1+e)/(1-e)\}\tan^2(\theta/2)}{1 + \{(1+e)/(1-e)\}\tan^2(\theta/2)}$$

From the given relation

$$= \frac{\{1 - \tan^2(\theta/2)\} - e\{1 + \tan^2(\theta/2)\}}{\{1 + \tan^2(\theta/2)\} - e\{1 - \tan^2(\theta/2)\}}$$

Divide above and below by $1 + \tan^2(\theta/2)$

$$\therefore \quad \cos\phi = \frac{\cos\theta - e}{1 - e\cos\theta}$$

Illustration 43

If $\cos\theta = \frac{a\cos\phi + b}{a + b\cos\phi}$, prove that $\tan(\theta/2) = \sqrt{[(a-b)/(a+b)]}\tan(\theta/2)$

Solution :

$$\cos\theta = \frac{a\cos\phi + b}{a + b\cos\phi} \qquad \therefore \quad \frac{1 - \tan^2(\theta/2)}{1 + \tan^2(\theta/2)}$$

$$= \frac{a\left\{1-\tan^2(\phi/2)\right\}+b\left\{1+\tan^2(\phi/2)\right\}}{a\left\{1+\tan^2(\phi/2)\right\}+b\left\{1-\tan^2(\phi/2)\right\}}$$

or $$\frac{1-\tan^2(\theta/2)}{1+\tan^2(\theta/2)} = \frac{(a+b)-(a-b)\tan^2(\phi/2)}{(a+b)+(a-b)\tan^2(\phi/2)}$$

Apply componendo and dividendo

$$\frac{2\tan^2(\theta/2)}{2} = \frac{2(a-b)\tan^2(\phi/2)}{2(a+b)}$$

$\therefore$ $$\tan(\theta/2) = \sqrt{(a-b)/(a+b)}\,\tan(\phi/2)$$

Illustration 44

If $\cos\theta = \dfrac{\cos\alpha - \cos\beta}{1-\cos\alpha\cos\beta}$, then prove that tan (θ/2) = ± tan (α/2) cot (β/2).

Solution :

$$\tan^2\frac{\theta}{2} = \frac{1-\cos\theta}{1+\cos\theta}$$

$$= \left(1-\frac{\cos\alpha-\cos\beta}{1-\cos\alpha\cos\beta}\right)-\left(1+\frac{\cos\alpha-\cos\beta}{1-\cos\alpha\cos\beta}\right)$$

$$= \frac{(1-\cos\alpha)+\cos\beta(1-\cos\alpha)}{(1+\cos\alpha)-\cos\beta(1+\cos\alpha)} = \frac{(1-\cos\alpha)(1+\cos\beta)}{(1+\cos\alpha)(1-\cos\beta)}$$

or $$\tan^2\frac{\theta}{2} = \frac{2\sin^2(\alpha/2)}{2\cos^2(\alpha/2)}\cdot\frac{2\cos^2(\beta/2)}{2\sin^2(\beta/2)}$$

$$= \tan^2(\alpha/2)\,.\,\cot^2(\beta/2)$$

$\therefore$ $$\tan(\theta/2) = \pm\tan(\alpha/2)\cot(\beta/2)$$

Illustration 45

If an angle θ be divided into two parts such that the tangent of one part is m times the tangent of the other, then prove that their difference ϕ is obtained from the equation sin ϕ = [(m – 1) / (m + 1)] sin θ.

Solution : Let the two parts be A and B so that

A + B = θ and A – B = ϕ and tan A = m tan B.

$$\therefore \qquad \frac{\tan A}{\tan B} = \frac{m}{1}$$

Apply componendo and dividendo

$$\therefore \qquad \frac{\tan A - \tan B}{\tan A + \tan B} = \frac{m-1}{m+1}$$

$$\text{or} \qquad \frac{\sin(A-B)}{\sin(A+B)} = \frac{m-1}{m+1}$$

or $\sin \phi = \{(m - 1)/(m + 1)\} \sin\theta.$

Illustration 46

If $\cos(\alpha + \beta) \sin(\gamma + \delta) = \cos(\alpha - \beta) \sin(\gamma - \delta)$, prove that $\cot \alpha \cot \beta \cot \gamma = \cot \delta$.

Solution :

From the given relation, we have

$$\frac{\sin(\gamma+\delta)}{\sin(\gamma-\delta)} = \frac{\cos(\alpha-\beta)}{\cos(\alpha+\beta)}$$

Now apply comp. and div.

$$\therefore \qquad \frac{\sin(\gamma+\delta)+\sin(\gamma-\delta)}{\sin(\gamma+\delta)-\sin(\gamma-\delta)} = \frac{\cos(\alpha-\beta)+\cos(\alpha+\beta)}{\cos(\alpha-\beta)-\cos(\alpha+\beta)}$$

$$\text{or} \qquad \frac{2\sin\gamma\cos\delta}{2\cos\gamma\sin\delta} = \frac{2\cos\alpha\cos\beta}{2\sin\alpha\sin\beta}$$

$\therefore \quad \cot \delta = \cot \alpha \cot \beta \cot \gamma.$

Illustration 47

If $\tan\frac{x-y}{2}$, $\tan z$, $\tan\frac{x+y}{2}$ are in G.P., then show that $\cos x = \cos y \cdot \cos 2z$.

Solution :

$$b^2 = ac \quad \Rightarrow \quad \frac{\sin^2 z}{\cos^2 z} = \frac{\sin\frac{x+y}{2}\sin\frac{x-y}{2}}{\cos\frac{x+y}{2}\cos\frac{x-y}{2}}$$

or $$\frac{\sin^2 z}{\cos^2 z} = \frac{\cos y - \cos x}{\cos y + \cos x}$$

Cross multiply or by C and D

$$\cos x.1 = \cos y\,(\cos^2 z - \sin^2 z) = \cos y \cos 2z.$$

Illustration 48

If $0 < \alpha, \beta, \gamma < \pi/2$, prove that $\sin\alpha + \sin\beta + \sin\gamma > \sin(\alpha + \beta + \gamma)$

Solution :

On putting the values of $\sin(\alpha + \beta + \gamma)$, we get $\sin\alpha + \sin\beta + \sin\gamma - \sin(\alpha + \beta + \gamma)$

$= \sin\alpha\,(1 - \cos\beta + \cos\gamma) + \sin\beta\,(1 - \cos\gamma\cos\alpha) + \sin\gamma\,(1 - \cos\alpha\cos\beta) + \sin\alpha\sin\beta\sin\gamma > 0$

$\therefore$ α, β, γ all lie in 1st quadrant.

$\therefore$ $\cos\alpha\cos\beta$ is +ive and less than 1 so that $1 - \cos\alpha\cos\beta$ is +ive and also $\sin\gamma$ is + ive. Thus every term in L.H.S. is + ive, i.e. L.H.S. > 0

$\therefore$ $\sin\alpha + \sin\beta + \sin\gamma > \sin(\alpha + \beta + \gamma)$

SUM OF SERIES

- **Arithmatic Series**

$$\sin a + \sin(a + d) + \sin(a + 2d) + \ldots + \sin(a + (n - 1)d) = \frac{\sin\frac{nd}{2}}{\sin\frac{d}{2}}\sin\left(\frac{2a + (n - 1)d}{2}\right)$$

$$\cos a + \cos(a + d) + \cos(a + 2d) + \ldots + \cos(a + (n - 1)d) = \frac{\sin\frac{nd}{2}}{\sin\frac{d}{2}}\cos\left(\frac{2a + (n - 1)d}{2}\right)$$

easier rule to remember : $$\frac{\sin\left(n.\frac{\text{difference}}{2}\right)}{\sin\left(\frac{\text{difference}}{2}\right)}\,(\sin)\text{ or }(\cos)\left(\frac{\text{first + last term}}{2}\right)$$

- **Geometric Series**

$$\cos A \cos 2A \cos 4A \ldots \cos 2^{n-1} A = \frac{\sin(2^n A)}{2^n \sin A}$$

Some more (which you can remember, if possible)

- $\sin\alpha + \sin\beta + \sin\gamma - \sin(\alpha + \beta + \gamma) = 4\sin\frac{\alpha+\beta}{2}\sin\frac{\beta+\gamma}{2}\sin\frac{\gamma+\alpha}{2}$

- $\cos\alpha + \cos\beta + \cos\gamma + \cos(\alpha + \beta + \gamma) = 4\cos\frac{\alpha+\beta}{2}\cos\frac{\beta+\gamma}{2}\cos\frac{\gamma+\alpha}{2}$

Illustration 49

For $0 < \phi \le \frac{\pi}{2}$, if $x = \sum_{n=0}^{\infty} \cos^{2n}\phi$, $y = \sum_{n=0}^{\infty} \sin^{2n}\phi$, $z = \sum_{n=0}^{\infty} \cos^{2n}\phi \sin^{2n}\phi$, then $x + y = xy$ and $xyz = x + y + z = xy + z$.

Solution :

All are infinite G.P.s with common ratio < 1.

$$x = \frac{1}{1-\cos^2\phi} = \frac{1}{\sin^2\phi},\ y = \frac{1}{1-\sin^2\phi} = \frac{1}{\cos^2\phi}$$

$$\therefore \quad x + y = \frac{1}{\sin^2\phi\cos^2\phi} = xy \qquad ...(1)$$

$$z = \frac{1}{1-\cos^2\phi\sin^2\phi} \qquad ...(2)$$

$$(x + y + z) = \left(\frac{1}{\sin^2\phi} + \frac{1}{\cos^2\phi}\right) + \frac{1}{1-\sin^2\phi\cos^2\phi}$$

$$= \frac{1}{\sin^2\phi\cos^2\phi} + \frac{1}{1-\sin^2\phi\cos^2\phi}$$

$$= \frac{1}{\sin^2\phi\cos^2\phi} \cdot \frac{1}{(1-\sin^2\phi\cos^2\phi)}$$

$= xyz$ by (1) and (2)

or $xy + z = xyz$ $\quad\therefore\quad x + y = xy$

Illustration 50

Show that $\cos^2 A + \cos^2 B - 2\cos A\cos B\cos(A + B) = \sin^2(A + B)$

Solution :

L.H.S. $= \cos^2 A + \cos^2 B - 2\cos A\cos B(\cos A\cos B - \sin A\sin B)$

$= (\cos^2 A - \cos^2 A \cos^2 B) + (\cos^2 B - \cos^2 A \cos^2 B) + 2 \sin A \sin B \cos A \cos B$

$= \cos^2 A (1 - \cos^2 B) + \cos^2 B (1 - \cos^2 A) + 2 \sin A \sin B \cos A \cos B$

$= (\sin A \cos B + \cos A \sin B)^2 = \sin^2 (A + B)$

Illustration 51

$$\frac{\sin x}{\cos 3x} + \frac{\sin 3x}{\cos 9x} + \frac{\sin 9x}{\cos 27x} = \frac{1}{2}(\tan 27x - \tan x)$$

Solution :

$2 = 3 - 1,\ 6 = 9 - 3,\ 18 = 27 - 9$

Use $\tan A - \tan B = \dfrac{\sin(A - B)}{\cos A \cos B}$

$$T_1 = \frac{1}{2}\left[\frac{2\sin x \cos x}{\cos 3x \cos x}\right] = \frac{1}{2}\frac{\sin(3x - x)}{\cos 3x \cos x}$$

$$= \frac{1}{2}[\tan 3x - \tan x]$$

$$T_2 = \frac{1}{2}[\tan 9x - \tan 3x],$$

$$T_3 = \frac{1}{2}[\tan 27x - \tan 9x] \text{ etc.}$$

Hence, $T_1 + T_2 + T_3 = \dfrac{1}{2}[\tan 27x - \tan x]$

Illustration 52

(a) For a positive integer n, **let** $f_n(\theta) = \left(\tan\dfrac{\theta}{2}\right)(1 + \sec\theta)(1 + \sec 2\theta)\ (1 + \sec 4\theta)\ldots(1 + \sec 2^n\theta)$

Then **(IIT 1999)**

(a) $f_2\left(\dfrac{\pi}{16}\right) = 1$ **(b)** $f_3\left(\dfrac{\pi}{32}\right) = 1$

(c) $f_4\left(\dfrac{\pi}{64}\right) = 1$ **(d)** $f_5\left(\dfrac{\pi}{128}\right) = 1$

Solution :

Ans. (a), (b), (c), (d).

$$E = f_n(\theta) = \frac{\sin(\theta/2)}{\cos(\theta/2)}\left[\frac{2\cos^2(\theta/2)}{\cos 2\theta}.\frac{2\cos^2\theta}{\cos 2\theta}.\frac{2\cos^2 2\theta}{\cos 4\theta}\ldots\right]$$

Combine first two factors

$$f_n(\theta) = \frac{\sin\theta}{\cos\theta}\left[\frac{2\cos^2}{\cos 2\theta}.\frac{2\cos^2 2\theta}{\cos 4\theta}\ldots\right]$$

Again combine first two factors

$= \tan 2\theta\ [\ ] = \tan 2^n\ \theta.$

$$\therefore\quad f_2\left(\frac{\pi}{16}\right) = \tan 4.\frac{\pi}{16} = \tan\frac{\pi}{4} = 1$$

$$\therefore\quad n = 2, \theta = \frac{\pi}{16}$$

Similarly each other is $\tan\frac{\pi}{4} = 1$.

Illustration 53

If the angle A of a triangle ABC is given by the equation 5 cos A + 3 = 0, then sin A and tan A are the roots of the equation $15x^2 + 8x - 16 = 0$

Solution :

We are given 5 cos A + 3 = 0 so that cos A = – 3/5 → A being the angle of ΔABC, sin A cannot be negative so cos A = – 3/5 ⇒ sin A = 4/5 and then tan A = – 4/3.

∴ sin A + tan A = – 8/15 = S

and sin A tan A = – 16/15 = P

∴ $x^2 - x\,S + P = 0$ is the required equation.

Illustration 54

If cosx + cosy + cosz = 0, then find the value of cos 3x + cos 3y + cos 3z.

Solution :

$$\sum\cos 3x = 4\sum\cos^3 x - 3\sum\cos x$$

$$= 4\sum a^3 - 0$$

Now if a + b + c = 0 then $a^3 + b^3 + c^3 = 3abc$.

$$\therefore\quad \sum\cos 3x = 4.3\ (\cos x\ \cos y\ \cos z)$$

Illustration 55

Prove :

(a) $\cos 5\theta = 16 \cos^5\theta - 20 \cos^3\theta + 5 \cos\theta$

(b) $\sin 5\theta = 5 \cos^4\theta \sin\theta - 10 \cos^2\theta \sin^3\theta + \sin^5\theta$

(c) Express $\sin 5\theta$ in terms of $\sin\theta$ and hence find the value of sin 36 .

Solution :

(a), (b), (c)

By De-Moivre's Theorem we know that

$(\cos\theta + i \sin\theta)^5 = \cos 5\theta + i \sin 5\theta$

L.H.S. on expansion by binomial theorem is

$\cos^5\theta + 5\cos^4\theta\ (i \sin\theta) + 10 \cos^3\theta\ (i \sin\theta)^2 + 10 \cos^2\theta\ (i \sin\theta)^3 + 5 \cos\theta\ (i \sin\theta)^4 + (i \sin\theta)^5$

Now equate real and imaginary parts and change

$\sin^2\theta$ to $1 - \cos^2\theta$ and $\cos^2\theta$ to $1 - \sin^2\theta$

depending on the answer

$\therefore \quad \cos 5\theta = \cos\theta\ (16\cos^4\theta - 20\cos^2\theta + 5)$ ...(1)

$\sin 5\theta = \sin\theta\ (16\sin^4\theta - 20\sin^2\theta + 5)$ ...(2)

Deduction :

If $\theta = 36$, then $5\theta = 180$

$\therefore \quad \sin 5\theta = 0$

Also $\sin 36 < \sin 45$ or $\sin^2 36 < \frac{1}{2}$

Now from (2) we get

$0 = s\ (16s^4 - 20s^2 + 5),\ s = \sin 36 \neq 0$

$$\therefore \quad s^2 = \frac{20 \pm \sqrt{400 - 320}}{32} = \frac{10 - 2\sqrt{5}}{16} \left(\because s^2 < \frac{1}{2}\right)$$

$$\therefore \quad s = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

Illustration 56

Prove the following :

$$\cos^3\theta + \cos^3 (120 + \theta) + \cos^3 (240 + \theta) = \frac{3}{4}\cos 3\theta$$

Solution :

We know that $\cos 3A = 4 \cos^3 A - 3 \cos A$

$$\therefore \quad \cos^3 A = \frac{1}{4}\ (\cos 3A + 3 \cos A)$$

Also $\cos(2n\pi + \theta) = \cos\theta$.

Applying the above we have

$$\text{L.H.S.} = \frac{1}{4}[(3\cos\theta + \cos 3\theta) + 3\cos(120 + \theta) + \cos(360 + 3\theta) + 3\cos(240 + \theta) + \cos(720 + 3\theta)$$

$$= \frac{1}{4}[(3\cos 3\theta] + \frac{3}{4}[\cos\theta + \cos(120 + \theta) + \cos(240 + \theta)]$$

$$= \frac{3}{4}\cos 3\theta + \frac{3}{4}[\cos\theta + 2\cos((180 + \theta)\cos 60]$$

$$= \frac{3}{4}\cos 3\theta + \frac{3}{4}\left[\cos\theta + 2.\frac{1}{2}(-\cos\theta)\right] = \frac{3}{4}\cos 3\theta.$$

Illustration 57

$$\cos^4\frac{\pi}{8} + \cos^4\frac{3\pi}{8} + \cos^4\frac{5\pi}{8} + \cos^4\frac{7\pi}{8} = \frac{3}{2}$$

Solution :

$\cos^4(7\pi/8) = \cos^4(\pi - -\pi/8) = \cos^4(\pi/8)$

$\cos^4(5\pi/8) = \cos^4(\pi/2 + \pi/8) = \sin^4(\pi/8)$

$\therefore$ L.H.S. $= 2[\cos^4 A + \sin^4 A]$, where $A = \pi/8$

$= 2[1 - 2\sin^2 A\cos^2 A] = 2 - [2\sin A\cos A]^2$

$$= 2 - \left(\sin 2.\frac{\pi}{8}\right)^2 = 2 - \sin^2\left(\frac{\pi}{4}\right) = 2 - \frac{1}{2} = \frac{3}{2}$$

Illustration 58

$$\tan^2\frac{\pi}{16} + \tan^2\frac{2\pi}{16} + \ldots + \tan^2\frac{6\pi}{16} + \tan^2\frac{7\pi}{16} = 35$$

Solution :

$$\left(\frac{\pi}{16} + \frac{7\pi}{16}\right) = \left(\frac{2\pi}{16} + \frac{6\pi}{16}\right) = \left(\frac{3\pi}{16} + \frac{5\pi}{16}\right) = \frac{\pi}{2}$$

Thus these are complementary.

and $\frac{4\pi}{16} = \frac{\pi}{4}$. If we take $\theta = \frac{\pi}{16}$

L.H.S. $= (\tan^2\theta + \cot^2\theta) + (\tan^2 2\theta + \cot^2 2\theta) + (\tan^2 3\theta + \cot^2 3\theta) + 1$

Now 1st bracket = $(\tan\theta + \cot\theta)^2 - 2$

$$= \left(\frac{1}{\sin\theta\cos\theta}\right)^2 - 2 = \left(\frac{2}{\sin 2\theta}\right)^2 - 2$$

$$= \frac{8}{1-\cos 4\theta} - 2 \text{ where } 4\theta = \frac{4\pi}{16} = \frac{\pi}{4}$$

$$= \frac{8\sqrt{2}}{\left(\sqrt{2}-1\right)} - 2 = 8\sqrt{2}\left(\sqrt{2}+1\right) - 2.$$

2nd bracket $= \dfrac{8}{1-\cos 8\theta} - 2 = 8 - 2 = 6$

3rd bracket $= \dfrac{8}{1-\cos 12\theta} = 8\sqrt{2}\left(\sqrt{2}-1\right) - 2$

and $\tan^2\dfrac{\pi}{4} = 1$

$\therefore$ S = 14 + 6 + 14 + 1 = 35.

Illustration 59

$\sum \sin^4 \dfrac{r\pi}{16} = \dfrac{3}{2}$, r = 1, 3, 5, 7

Solution :

$$\left(\frac{7\pi}{16} + \frac{\pi}{16}\right) = \frac{\pi}{2}, \left(\frac{5\pi}{16} + \frac{3\pi}{16}\right) = \frac{\pi}{2}$$

Hence by C.R.,

$(\sin^4 x + \cos^4 x) + (\sin^4 y + \cos^4 y)$

$= 1 - 2\sin^2 x\cos^2 x + 1 - 2\sin^2 y\cos^2 y$

$$= 2 - \frac{1}{2}[\sin^2 2x + \sin^2 2y] = 2 - \frac{1}{2}\left[\sin^2\frac{\pi}{8} + \sin^2\frac{3\pi}{8}\right]$$

$$= 2 - \frac{1}{2}\left[\sin^2\frac{\pi}{8} + \cos^2\frac{\pi}{8}\right] = \frac{3}{2} \text{ by C.R.}$$

Illustration 60

$\tan\theta \tan(\theta + 60) + \tan\theta \tan(\theta - 60) + \tan(\theta + 60)\tan(\theta - 60) = -3.$

Solution :

We know that $\tan A \tan B + 1 = \left(\dfrac{\sin A \sin B}{\cos A \cos B} + 1\right)$

$$= \frac{\cos A \cos B + \sin A \sin B}{\cos A \cos B} = \frac{\cos(A - B)}{\cos A \cos B}$$

Now take – 3 from R.H.S. to L.H.S. and write it as 1 + 1 + 1.

Hence we have to prove that

$[\tan\theta \tan(\theta - 60) + 1] + [\tan\theta \tan(\theta - 60) + 1] + [\tan(\theta + 60)\tan(\theta - 60) + 1] = 0$

or $$\frac{\cos(\theta - \theta - 60°)}{\cos\theta \cos(\theta + 60°)} + \frac{\cos(\theta - \theta + 60°)}{\cos\theta \cos(\theta - 60°)} + \frac{\cos(\theta + 60° - \theta + 60°)}{\cos(\theta + 60°)\cos(\theta - 60°)}$$

$$= \frac{\cos 60° \cos(\theta - 60°) + \cos 60° \cos(\theta + 60°) + \cos 120° \cos\theta}{\cos\theta \cos(\theta + 60°)\cos(\theta - 60°)}$$

$$= \frac{\frac{1}{2}[\cos(\theta - 60°) + \cos(\theta + 60°)] + \cos(90° + 30°)\cos\theta}{D^r}$$

$$= \frac{\frac{1}{2}.2\cos\theta \cos 60° - \sin 30° \cos\theta}{D^r} = \frac{1}{2}\frac{(\cos\theta - \cos\theta)}{D^r} = 0$$

Illustration 61

If $x \sin\theta = y \sin\left(\theta + \dfrac{2\pi}{3}\right) = z \sin\left(\theta + \dfrac{4\pi}{3}\right)$ then prove that $\Sigma xy = 0$

Solution :

Let each be equal to 1/k.

We have to prove that $xy + yz + zx = 0$

or $\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} = 0,$ on dividing by xyz

On putting the values $\sum \dfrac{1}{x} = k\left[\sin\theta + \sin\left(\theta + \dfrac{2\pi}{3}\right) + \sin\left(\theta + \dfrac{4\pi}{3}\right)\right] = 0$

Combine Ist and 3rd

$$= k\left[2\sin\left(\theta+\frac{2\pi}{3}\right)\cos\frac{2\pi}{3}+\sin\left(\theta+\frac{2\pi}{3}\right)=0\right]$$

$$\therefore \quad 2\cos\frac{2\pi}{3}=2\left(-\frac{1}{2}\right)=-1$$

$$\therefore \quad \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=0 \qquad \text{or} \qquad xy + yz + zx = 0$$

Illustration 62

(a) $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}=-\frac{1}{2}$

(b) $\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}=\frac{1}{2}$

(c) $\cos 0+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}=1$

Solution :

(a) angles being in A.P. of three terms $\alpha = 2\pi/7 = \beta$, $n = 3$, $\beta/2 = \pi/7$

$$S=\frac{\sin(3\pi/7)}{\sin(\pi/7)}\cos\frac{1}{2}\left(\frac{2\pi}{7}+\frac{6\pi}{7}\right)$$

$$=\frac{\sin(4\pi/7)}{\sin(\pi/7)}\cos\frac{4\pi}{7}$$

$$=\frac{1}{2}\frac{\sin(8\pi/7)}{\sin(\pi/7)}=\frac{1}{2}\frac{\sin(\pi+\pi/7)}{\sin(\pi/7)}=-\frac{1}{2}.$$

(b) Here the angles are in A.P. of 5 terms.

$$n=5,\ \alpha=\frac{\pi}{11},\ \beta=\frac{2\pi}{11} \qquad \therefore \qquad \frac{\beta}{2}=\frac{\pi}{11}$$

$$S=\frac{\sin 5.\frac{\pi}{11}}{\sin\frac{\pi}{11}}.\cos\frac{1}{2}\left[\frac{\pi}{11}+\frac{9\pi}{11}\right]$$

$$=\frac{2\sin\frac{5\pi}{11}\cos\frac{5\pi}{11}}{2\sin\frac{\pi}{11}}=\frac{\sin\frac{10\pi}{11}}{2\sin\frac{\pi}{11}}\sin\left(\pi-\frac{\pi}{11}\right)=\frac{1}{2}$$

(c) **Hint :** $\cos 0 = 1$, $\cos\frac{6\pi}{7} = -\cos\frac{\pi}{7}$,

Now solve it yourself.

Illustration 63

(a) $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14} = \frac{1}{8}$

(b) $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\sin\frac{9\pi}{14}\sin\frac{11\pi}{14}\sin\frac{13\pi}{14} = \frac{1}{64}$

Solution :

(a) $\sin\frac{\pi}{14} = \sin\left(\frac{\pi}{2} - \frac{6\pi}{14}\right) = \cos\frac{6\pi}{14}$

$= \cos\left(\pi - \frac{8\pi}{14}\right) = -\cos\frac{8\pi}{14}$

$\sin\frac{3\pi}{14} = \sin\left(\frac{\pi}{2} - \frac{4\pi}{14}\right) = \cos\frac{4\pi}{14}$

$\sin\frac{5\pi}{14} = \sin\left(\frac{\pi}{2} - \frac{2\pi}{14}\right) = \cos\frac{2\pi}{14}$

$\therefore$ L.H.S. $= -\cos\frac{2\pi}{14}\cos\frac{4\pi}{14}\cos\frac{8\pi}{14}$

$= -\frac{1}{2^3\sin A}.\sin\left(2^3 A\right), A = \frac{2\pi}{14}$

$= -\frac{1}{8\sin\frac{\pi}{7}}\sin\frac{8\pi}{7} = -\frac{1}{8\sin\frac{\pi}{7}}\sin\left(\pi + \frac{\pi}{7}\right)$

$= -\frac{1}{8}(-1) = \frac{1}{8}$

$\therefore$ $\sin(\pi + \theta) = -\sin\theta$

(b) $\sin\frac{7\pi}{14} = \sin\frac{\pi}{2} = 1,$

$$\sin\frac{13\pi}{14} = \sin\left(\pi - \frac{\pi}{14}\right) = \sin\frac{\pi}{14}$$

Similarly $\sin\frac{11\pi}{14} = \sin\frac{3\pi}{14}$ and $\sin\frac{9\pi}{14} = \sin\frac{5\pi}{14}$

Hence the question reduces to square of part (a)

$\therefore$ Ans. $= \left(\frac{1}{8}\right)^2 = \frac{1}{64}$

Illustration 64

sin 20 sin 40 sin 60 sin 80 = $\frac{3}{16}$

Solution :

L.H.S. $= \left(\sqrt{3}/2\right)\sin 20° \sin(60° - 20°) \sin(60° + 20°)$

$= \left(\sqrt{3}/2\right)\sin 20° (\sin^2 60° - \sin^2 20°)$

$= \left(\sqrt{3}/2\right)\sin 20°[(3/4) - \sin^2 20°]$

$= \left(\sqrt{3}/8\right)[3\sin 20° - 4\sin^3 20°]$

$= \left(\sqrt{3}/8\right)\sin(3.20°)$

$= \left(\sqrt{3/8}\right)\sin 60° = \left(\sqrt{3/8}\right).\left(\sqrt{3/2}\right) = 3/16.$

Illustration 65

$(2\cos\theta - 1)(2\cos 2\theta - 1)(2\cos 2^2\theta - 1) \ldots (2\cos 2^{n-1}\theta - 1) = \frac{2\cos 2^n\theta + 1}{2\cos\theta + 1}$

Solution :

Try it yourself.

Illustration 66

$$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)=\frac{1}{8}$$

Solution :

By supplementary rule

$\cos(\pi - A) = -\cos A$

$$\text{L.H.S.} = \left(1+\cos\frac{\pi}{10}\right)\left(1+\cos\frac{3\pi}{10}\right)\left(1-\cos\frac{3\pi}{10}\right)\left(1-\cos\frac{\pi}{10}\right)$$

$$= \left(1-\cos^2\frac{\pi}{10}\right)\left(1-\cos^2\frac{3\pi}{10}\right) = \sin^2 18^\circ \sin^2 54^\circ$$

$$= \left(\frac{\sqrt{5}-1}{4}.\frac{\sqrt{5}+1}{4}\right)^2 = \frac{1}{16}$$

Illustration 67

If $\cos(\alpha+\beta) = (4/5)$ and $\sin(\alpha-\beta) = (5/13)$ and α, β lie between 0 and $\pi/4$ find $\tan 2\alpha$.

Solution :

$\cos(\alpha+\beta) = 4/5$

$\therefore \quad \tan(\alpha+\beta) = \sqrt{(5^2-4^2)} - 4 = 3/4$

$\sin(\alpha-\beta) = 5/13$

$\therefore \quad \tan(\alpha-\beta) = 5/\sqrt{(13^2-5^2)} = 5/12$

$\tan 2\alpha = \tan(\alpha+\beta+\alpha-\beta)$

$$= \frac{\tan(\alpha+\beta)+\tan(\alpha-\beta)}{1-\tan(\alpha+\beta)\tan(\alpha-\beta)} = \frac{3/4+5/12}{1-3/4.5/12} = \frac{56}{33}$$

Illustration 68

(a) If $\sin x + \cos x = a$, evaluate $\sin^6 x + \cos^6 x$.

(b) If $\sin A + \sin 2A = x$, $\cos A + \cos 2A = y$, then prove that $(x^2+y^2)(x^2+y^2-3) = 2y$.

Solution :

(a) $\sin^6 x + \cos^6 x$

$$= 1 - 3\sin^2 x\cos^2 x = \frac{1}{4}[4 - 3\sin^2 2x] \qquad ...(1)$$

Now $\sin x + \cos x = a$. Squaring

$1 + \sin 2x = a^2 \qquad \therefore \qquad \sin 2x = a^2 - 1$

Putting in (1),

$\therefore$ L.H.S. $= \frac{1}{4}\,[4 - 3\,(a^2 - 1)^2]$.

$-1 \le \sin 2x \le 1 \quad \Rightarrow \quad 1 \le a^2 - 1 \le 1$

or $\quad 0 \le a^2 \le 2 \qquad \therefore \qquad a^2 \ge 0$ and $a^2 \le 2$

which implies that $-\sqrt{2} \le a \le \sqrt{2}$.

(b) Squaring and adding the given relations, we get

$1 + 1 + 2\cos(2A - A) = x^2 + y^2$

$$\therefore \quad \cos A = \frac{1}{2}(x^2 + y^2 - 2) \qquad ...(1)$$

Now from second relation, we get

$\cos A + 2\cos^2 A - 1 = y$

or $\quad 2\cos^2 A + \cos A - 1 = y$

or $\quad (\cos A + 1)(2\cos A - 1) = y$

Now put for cos A from (1).

Illustration 69

If x and y are acute angles such that x + y and x – y satisfy the equation $\tan^2\theta - 4\tan\theta + 1 = 0$, then x = ..., y = ...

Solution : $x = \pi/4$, $y = \pi/6$

The roots of quadratic in $\tan\theta$ are

$\tan(x + y)$ and $\tan(x - y)$

$\therefore \quad S = \tan(x + y) + \tan(x - y) = 4 \qquad ...(1)$

and $\quad P = 1$

$\therefore \quad \tan 2x = \tan(x + y + x - y)$

$$= \frac{S}{1 - P} = \infty = \tan\frac{\pi}{2}$$

$$\therefore \quad 2x = \frac{\pi}{2} \quad \text{or} \quad x = \frac{\pi}{4}$$

Now put $x = \frac{\pi}{4}$ in S = 4 i.e. in (1)

$$\therefore \quad \frac{1+t}{1-t} + \frac{1-t}{1+t} = 4 \quad \text{where } t = \tan y$$

or $\frac{2(1+t^2)}{1-t^2} = 4$ $\therefore$ $\cos 2y = \frac{1}{2}$

$\therefore$ $2y = \pi/3$ or $y = \pi/6$

Illustration 70

Show that for all real values of θ, the expression $a \sin^2\theta + b \sin\theta \cos\theta + c \cos^2\theta$ lies between $\frac{1}{2}(a+c) - \frac{1}{2}\sqrt{b^2+(a-c)^2}$ and $\frac{1}{2}(a+c) + \frac{1}{2}\sqrt{b^2+(a-c)^2}$.

Solution :

The given expression

$$= \frac{1}{2}[a(1-\cos 2\theta) + b\sin 2\theta + c(1+\cos 2\theta)]$$

$$= \frac{1}{2}(a+c) + \frac{1}{2}b\sin 2\theta + \frac{1}{2}(c-a)\cos 2\theta$$

Hence by the least and greatest values of the given expression are respectively

$$\frac{1}{2}(a+c) - \frac{1}{2}\sqrt{b^2+(c-a)^2}$$

and $$\frac{1}{2}(a+c) + \frac{1}{2}\sqrt{\left\{b^2+(c-a)^2\right\}}$$

$$\left\{\text{as } -\sqrt{a^2+b^2} \le a\sin\theta + b\cos\theta \le \sqrt{a^2+b^2}\right\}$$

so that the value of the expression lies between these values as required.

Illustration 71

If $\cos^2\theta = \frac{1}{2}(a^2-1)$ and $\tan^2\left(\frac{\theta}{2}\right) = \tan^{2/3}\alpha$, prove that $\cos^{2/3}\alpha + \sin^{2/3}\alpha = (2/a)^{2/3}$.

Solution :

We have $\tan^{2/3}\alpha = \tan^2\frac{\theta}{2} = \frac{1-\cos\theta}{1+\cos\theta} = \frac{\sqrt{3}-\sqrt{(a^2-1)}}{\sqrt{(3)}+\sqrt{(a^2-1)}}$

$$\therefore \quad \frac{\sin^{2/3}\alpha}{\sqrt{(3)}-\sqrt{(a^2-1)}} = \frac{\cos^{2/3}}{\sqrt{(3)}+\sqrt{(a^2-1)}}$$

or $\frac{a}{p} = \frac{b}{q}$

Now apply the rule of ratio proportion

$$\frac{a}{p} = \frac{b}{q} = \frac{a+b}{p+q} = \left(\frac{a^3 + b^3}{p^3 + q^3}\right)^{1/3} \qquad ...(2)$$

$p + q = 2\sqrt{3}$

$p^3 + q^3 = 2[3\sqrt{3} + 3\sqrt{3}(a^2 - 1)] = 6\sqrt{3}\,a^2$

$a^3 + b^3 = \sin^2\alpha + \cos^2\alpha = 1,$

$a + b = \sin^{2/3}\alpha + \cos^{2/3}\alpha$

$\therefore$ From last two relations of (2), we have

$\sin^{2/3}\alpha + \cos^{2/3}\alpha$

$$= 2\sqrt{3}\left[\frac{1}{6\sqrt{3}\,a^2}\right]^{1/3} = \left[\frac{24\sqrt{3}}{6\sqrt{3}\,a^2}\right]^{1/3}$$

$= (4/a^2)^{1/3} = (2/a)^{2/3}.$

Illustration 72

(a) If $\dfrac{\tan(\alpha+\beta-\gamma)}{\tan(\alpha-\beta+\gamma)} = \dfrac{\tan\gamma}{\tan\beta}$, then prove that either $\sin(\beta-\gamma) = 0$ or $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0$.

(b) If $\cos(\beta-\gamma) + \cos(\gamma-\alpha) + \cos(\alpha-\beta) = -3/2$, then prove that $\Sigma\cos\alpha = 0$, $\Sigma\sin\alpha = 0$

Solution :

(a) Apply componendo and dividendo and remember

$$\tan A \pm \tan B = \frac{\sin(A \pm B)}{\cos A \cos B}$$

$$\therefore \quad \frac{\sin 2\alpha}{\sin 2(\beta-\gamma)} = \frac{\sin(\beta+\gamma)}{-\sin(\beta-\gamma)}$$

or $\sin(\beta-\gamma)\,[\sin 2\alpha + 2\cos(\beta-\gamma)\sin(\beta+\gamma)] = 0$

or $\sin(\beta-\gamma)\,[\sin 2\alpha + \sin 2\beta + \sin 2\gamma] = 0$

(b) $3 = 1 + 1 + 1 = (\cos^2\alpha + \sin^2\alpha) + (\ ) + (\ )$

or $(\cos^2\alpha + \cos^2\beta + \cos^2\gamma + 2\cos\beta\cos\gamma + 2\cos\gamma\cos\alpha + 2\cos\alpha\cos\beta\ ) + (\text{In sin}) = 0$

or $(\cos\alpha + \cos\beta + \cos\gamma)^2 + (\ )^2 = 0$

$\therefore \quad \Sigma\cos\alpha = 0,\ \Sigma\sin\alpha = 0$

$[\therefore\ x^2 + y^2 = 0 \Rightarrow x = 0,\ y = 0]$

Illustration 73

Prove that tan 18 is a root of the equation $5x^4 - 10x^2 + 1 = 0$ and hence prove that $\tan^2 18 = 0.1056$

Solution :

If $\theta = 18$ then $5\theta = 90$ $\therefore$ $3\theta = 90 - 2\theta$

or $\tan 3\theta = \cot 2\theta$ or $\dfrac{3t - t^3}{1 - 3t^2} = \dfrac{1 - t^2}{2t}$

$\therefore$ $5t^4 - 10t^2 + 1 = 0$ $\therefore$ $t^2 = 1 \pm \sqrt{0.8}$

or $t^2 = 1 \pm 0.8944 = 1 - .8944 = 0.1056$

$t^2 = \tan^2 18$ is +ive and less than 1.

$\therefore$ $\tan 45 = 1$ $\therefore$ $\tan 18 < \tan 45$ or $t < 1$

and +ive. The other value being > 1 is rejected.

Illustration 74

Determine the smallest positive value of x (in degrees) for which

$\tan (x + 100) = \tan (x + 50) \tan (x) \tan (x - 50)$

Solution :

From the given relation, we have

$$\frac{\tan(x + 100°)}{\tan(x - 50°)} = \tan(x + 50°) \tan x$$

We have chosen the combination as above because sum of the angles in both sides is $(2x + 50)$.

$$\frac{\sin(x + 100°)\cos(x - 50°)}{\cos(x + 100°)\sin(x - 50°)} = \frac{\sin(x + 50°)\sin x}{\cos(x + 50°)\cos x}$$

Apply componendo and dividendo

$$\frac{\sin(2x + 50°)}{\sin 150°} = \frac{\cos 50°}{-(\cos 2x + 50°)}$$

$2 \sin (2x + 50) \cos (2x + 50) = - \cos 50$

$\therefore$ $\sin 150 = \sin 30 = \dfrac{1}{2}$

$\sin (4x + 100) = - \sin 40$

$= \sin (180 + 40) = \sin 220$

or $= \sin (2\pi - 40) = \sin 320$

$\therefore$ $4x + 100 = 220$ or 320

or $4x = 120$ or 220 or $x = 30$ or 55

Illustration 75

If $\dfrac{\cos^4 x}{\cos^2 y} + \dfrac{\sin^4 x}{\sin^2 y} = 1$, prove that $\dfrac{\cos^4 y}{\cos^2 x} + \dfrac{\sin^4 y}{\sin^2 x} = 1$.

Solution :

A careful look at the question suggests that we have to prove x = y. We know that

$2\cos^2\theta = 1 + \cos 2\theta$,

and $2\sin^2\theta = 1 - \cos 2\theta$.

Hence changing to double angles in the given relation

$$\frac{\cos^4 x}{\cos^2 y} + \frac{\sin^4 x}{\sin^2 y} = 1, \text{ we get}$$

$$\frac{1}{4}(1+\cos 2x)^2 \frac{1}{2}(1-\cos 2y) + \frac{1}{4}(1-\cos 2x)^2 . \frac{1}{2}(1+\cos 2y)$$

$$= \frac{1}{2}(1+\cos 2y).\frac{1}{2}(1-\cos 2y)$$

or $\left\{(1+\cos 2x)^2 + (1-\cos 2x)^2\right\} - \cos 2y\left\{(1+\cos 2x)^2 - (1-\cos 2x)^2\right\}$

$= 2(1 - \cos^2 2y)$

or $2(1 + \cos^2 2x) - \cos 2y\,(4\cos 2x) = 2 - 2\cos^2 2y$

or $\cos^2 2x + \cos^2 2y - 2\cos 2x \cos 2y = 0$

or $(\cos 2x - \cos 2y)^2 = 0$

$\therefore$ $\cos 2x = \cos 2y$ or $y = n\pi + x$

$$\therefore \quad \frac{\cos^4 y}{\cos^2 x} + \frac{\sin^4 y}{\sin^2 x} = \cos^2 x + \sin^2 x = 1$$

(on putting y = x)

Illustration 76

If $\dfrac{\sin^4 A}{a} + \dfrac{\cos^4 A}{b} = \dfrac{1}{a+b}$, then prove that $\dfrac{\sin^8 A}{a^3} + \dfrac{\cos^8 A}{b^3} = \dfrac{1}{(a+b)^3}$.

Solution :

$(a + b)(b\sin^4 A + a\cos^4 A) - ab = 0$

or $ab[\sin^4 A + \cos^4 A - 1] + a^2\cos^4 A + b^2\sin^4 A = 0$

or $ab[1 - 2\sin^2 A\cos^2 A - 1] + a^2\cos^4 A + b^2\sin^4 A = 0$

or $(a\cos^2 A - b\sin^2 A)^2 = 0$

or $\dfrac{\cos^2 A}{b} = \dfrac{\sin^2 A}{a} = \dfrac{1}{a+b}$ ...(1)

$\therefore \quad \dfrac{\sin^8 A}{a^3} + \dfrac{\cos^8 A}{b^3} = \dfrac{1}{a^3} \cdot \dfrac{a^4}{(a+b)^4} + \dfrac{b^4}{(a+b)^4}$ by (1)

$= \dfrac{(a+b)}{(a+b)^4} = \dfrac{1}{(a+b)^3}$

Illustration 77

(a) Sum the following series :

$$\frac{1}{\sin\theta \sin 2\theta} + \frac{1}{\sin 2\theta \sin 3\theta} + \frac{1}{\sin 3\theta \sin 4\theta} + \ldots n \text{ terms.}$$

(b) sec θ. sec 2θ + sec 2θ sec 3θ + ... n terms

Solution :

(a) The difference of the angles is θ.

Multiply above and below by sin θ.

$$S = \frac{1}{\sin\theta}\left[\frac{\sin\theta}{\sin\theta \sin 2\theta} + \frac{\sin\theta}{\sin 2\theta \sin 3\theta} + \ldots\right]$$

$$T_1 = \frac{\sin(2\theta - \theta)}{\sin\theta \sin 2\theta} = \cot\theta - \cot 2\theta$$

$$S_n = \frac{1}{\sin\theta}[(\cot\theta - \cot 2\theta) + (\cot 2\theta - \cot 3\theta) + \ldots + \{\cot n\theta - \cot(n+1)\theta\}]$$

$$= \frac{1}{\sin\theta}[\cot\theta - \cot(n+1)\theta]$$

(b) Change in cos and proceed as in part (a) by multiplying above and below by sin θ.

$$\therefore \quad S = \frac{1}{\sin\theta}[\tan(n+1)\theta - \tan n\theta]$$

The angles of triangle and related identities

If A + B + C = π, then :

- sin 2A + sin 2B + sin 2C = 4 sinA sinB sinC
- cos 2A + cos 2B + cos 2C = –1 – 4 cosA cosB cosC
- cosA + cosB + cosC = 1 + 4 sinA/2 sinB/2 sinC/2
- sinA + sinB + sinC = 4 cosA/2 cosB/2 cosC/2
- tanA + tanB + tanC = tanA tanB tanC
- cotA cotB + cotB cotC + cotC cotA = 1
- tanA/2 tanB/2 + tanB/2 tanC/2 + tanC/2 tanA/2 = 1
- cotA/2 + cotB/2 + cotC/2 = cotA/2 cotB/2 cotC/2

Illustration 78

Prove the following for A + B + C = π

(a) sin 2A + sin 2B + sin 2C = 4 sinA sinB sinC

(b) cos 2A + cos 2B + cos 2C = –1 – 4 cosA cosB cosC

(c) $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{A}{2}\cos\frac{C}{2}$

Solution :

(a) L.H.S. = 2 sin A cos A + 2 sin (B + C) cos (B – C)
= 2 sin A cos A + 2 sin A cos (B – C)
= 2 sin A [cos A + cos (B – C)]
= 2 sin A [cos (B – C) – cos (B + C)] [∴ cos A = – cos (B + C)]
= 2 sin A . 2 sin B sin C
= 4 sin A sin B sin C.

(b) In the answer we want –1 and as such we write cos 2A as 2 $\cos^2$ A – 1 and combine the other two terms.

L.H.S = (2 $\cos^2$ A – 1) + 2 cos (B + C) cos (B – C)
= –1 + 2 $\cos^2$ A – 2 cos A cos (B – C)
= –1 + 2 cos A [cos A – cos (B – C)]
= –1 + 2 cos A [–cos (B + C) –cos (B – C)]
= –1 – 2 cos A (2 cos B cos C)
= –1 – 4 cos A cos B cos C.

(c) L.H.S. = 2 sin (A/2) cos (A/2) + 2 sin {(B + C)/2} cos {(B – C)/2}
= 2 sin (A/2) cos (A/2) + 2 cos (A/2) cos {(B – C)/2}
= 2 cos (A/2) [sin(A/2) + cos {(B – C)/2}]
= 2 cos (A/2) [cos {(B + C)/2} + cos {(B – C)/2}]
= 2 cos (A/2) [2 cos (B/2) cos (C/2)
= 4 cos (A/2) cos (B/2) cos (C/2)

Illustration 79

If $\alpha + \beta + \gamma = \pi$, prove that

$\sin^2\alpha + \sin^2\beta - \sin^2\gamma = 2\sin\alpha \sin\beta \cos\gamma$ **[IIT–1980]**

Solution :

We are given $\alpha + \beta + \gamma = \pi$,

$\Rightarrow \quad \alpha + \beta = \pi - \gamma$

$\Rightarrow \quad \cos(\alpha + \beta) = \cos(\pi - \gamma)$

or $\quad \cos\alpha \cos\beta - \sin\alpha \sin\beta = -\cos\gamma \quad [\because \cos(\pi - \gamma) = -\cos\gamma]$

or $\quad \sin\alpha \sin\beta - \cos\gamma = \cos\alpha \cos\beta$

[Note that we have brought $\sin\alpha \sin\beta$ and $\cos\gamma$ on one side because term without square in the given identity is $2\sin\alpha \sin\beta \cos\gamma$]

Squaring, we get

$(\sin\alpha \sin\beta - \cos\gamma)^2 = \cos^2\alpha \cos^2\beta$

or $\quad \sin^2\alpha \sin^2\beta + \cos^2\gamma - 2\sin\alpha \sin\beta \cos\gamma = (1 - \sin^2\alpha)(1 - \sin^2\beta)$

or $\quad \sin^2\alpha \sin^2\beta + \cos^2\gamma - 2\sin\alpha \sin\beta \cos\gamma$

$= 1 - \sin^2\alpha - \sin^2\beta + \sin^2\alpha \sin^2\beta$

or $\quad \sin^2\alpha + \sin^2\beta + \cos^2\gamma - 1 = 2\sin\alpha \sin\beta \cos\gamma$

or $\quad \sin^2\alpha + \sin^2\beta + 1 - \sin^2\gamma - 1 = 2\sin\alpha \sin\beta \cos\gamma$

or $\quad \sin^2\alpha + \sin^2\beta - \sin^2\gamma = 2\sin\alpha \sin\beta \cos\gamma$

Illustration 80

If A + B + C = 180 , Prove that

$$\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} = 1 - 2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

Solution :

Given $\quad A + B\ C = 180$

$$\therefore \quad \frac{A}{2} + \frac{B}{2} + \frac{C}{2} = 90^\circ \quad \text{or, } \frac{A}{2} + \frac{B}{2} = 90^\circ - \frac{C}{2}$$

$$\therefore \quad \cos\left(\frac{A}{2} + \frac{B}{2}\right) = \cos\left(90^\circ - \frac{C}{2}\right)$$

$$\text{or,} \quad \cos\frac{A}{2}\cos\frac{B}{2} - \sin\frac{A}{2}\sin\frac{B}{2} = \sin\frac{C}{2}$$

$$\text{or,} \quad \sin\frac{A}{2}\sin\frac{B}{2} + \sin\frac{C}{2} = \cos\frac{A}{2}\cos\frac{B}{2}$$

[Note that $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ occur without square in the given identity, therefore, we have brought $\sin\frac{A}{2}\sin\frac{B}{2}$ and $\sin\frac{C}{2}$ on one side.]

or, $\left(\sin\frac{A}{2}\sin\frac{B}{2}+\sin\frac{C}{2}\right)^2=\cos^2\frac{A}{2}\cos^2\frac{B}{2}$

or, $\sin^2\frac{A}{2}\sin^2\frac{B}{2}+\sin^2\frac{C}{2}+2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$=\left(1-\sin^2\frac{A}{2}\right)\left(1-\sin^2\frac{B}{2}\right)$

or, $\sin^2\frac{A}{2}\sin^2\frac{B}{2}+\sin^2\frac{C}{2}+2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$=1-\sin^2\frac{A}{2}-\sin^2\frac{B}{2}+\sin^2\frac{A}{2}\sin^2\frac{B}{2}$

or, $\sin^2\frac{C}{2}+2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}=1-\sin^2\frac{A}{2}-\sin^2\frac{B}{2}$

or, $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2}=1-2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

Illustration 81

Show that $\sin^2\alpha+\sin^2\beta+2\sin\alpha\sin\beta\cos(\alpha+\beta)=\sin^2(\alpha+\beta)$.

Solution :

[Here three angles α, β and $\alpha+\beta$ occur and the term without square is $2\sin\alpha\sin\beta\cos(\alpha+\beta)$ which has a factor $\sin\alpha\sin\beta$, therefore we will keep α and β one one side.]

Let $\alpha+\beta=\gamma$, then $\cos(\alpha+\beta)=\cos\gamma$

or, $\cos\alpha\cos\beta-\sin\alpha\sin\beta=\cos\gamma$

or, $\sin\alpha\sin\beta+\cos\gamma=\cos\alpha\cos\beta$

or, $(\sin\alpha\sin\beta+\cos\gamma)^2=\cos^2\alpha\cos^2\beta$

$=(1-\sin^2\alpha)(1-\sin^2\beta)$

or, $\sin^2\alpha\sin^2\beta+\cos^2\gamma+2\sin\alpha\sin\beta\cos\gamma=1-\sin^2\alpha-\sin^2\beta+\sin^2\alpha\sin^2\beta$

or, $\cos^2\gamma+2\sin\alpha\sin\beta\cos\gamma=1-\sin^2\alpha-\sin^2\beta$

or, $\sin^2\alpha+\sin^2\beta+2\sin\alpha\sin\beta\cos\gamma=1-\cos^2\gamma=\sin^2\gamma$

$\therefore$ $\sin^2\alpha+\sin^2\beta+2\sin\alpha\sin\beta\cos(\alpha+\beta)=\sin^2(\alpha+\beta)$ [$\because \gamma=\alpha+\beta$]

Illustration 82

In a ΔABC, prove that tanA + tanB + tanC = tanA tanB tanC

Solution :

In ΔABC, $A + B + C = \pi$ $\quad A + B = \pi - C$

or, $\tan(A + B) = \tan(\pi - C)$

or, $\dfrac{\tan A + \tan B}{1 - \tan A \tan B} = \tan C$ $\quad [\because \tan(\pi - C) = -\tan C]$

or, $\tan A + \tan B = -\tan C + \tan A \tan B \tan C$

or, $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

Illustration 83

If A + B + C = π, prove that

$$\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$$

Solution :

$\because \quad A + B + C = \pi$

$\because \quad \dfrac{A}{2} + \dfrac{B}{2} + \dfrac{C}{2} = \dfrac{\pi}{2}$ or, $\dfrac{A}{2} + \dfrac{B}{2} = \dfrac{\pi}{2} - \dfrac{C}{2}$

$\because \quad \tan\left(\dfrac{A}{2} + \dfrac{B}{2}\right) = \tan\left(\dfrac{\pi}{2} - \dfrac{C}{2}\right) = \cot\dfrac{C}{2}$

or, $\dfrac{\tan\frac{A}{2} + \tan\frac{B}{2}}{1 - \tan\frac{A}{2}\tan\frac{B}{2}} = \dfrac{1}{\tan\frac{C}{2}}$

or, $\tan\dfrac{A}{2}\tan\dfrac{C}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} = 1 - \tan\dfrac{A}{2}\tan\dfrac{B}{2}$

or, $\tan\dfrac{A}{2}\tan\dfrac{B}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} + \tan\dfrac{C}{2}\tan\dfrac{A}{2} = 1.$

Illustration 84

In a ΔABC, if cotA + cotB + cotC = $\sqrt{3}$. prove that the triangle is equilateral.

Solution : In ΔABC, $A + B + C = \pi$

show that

$\cot A \cot B + \text{Cot}B \cot C + \cot C \cot A = 1$ ...(1)

as an exercise for the student

Again, given $\cot A + \cot B + \cot C = \sqrt{3}$

or, $(\cot A + \cot B + \cot C)^2 = 3$

or, $\cot^2 A + \cot^2 B + \cot^2 C + (\cot A \cot B + \cot B \cot C + \cot C \cot A) = 3$

or, $\cot^2 A + \cot^2 B + \cot^2 C + 2 \quad 1 = 3$ [from (1)]

or, $\cot^2 A + \cot^2 B + \cot^2 C = 3 \quad 2 = 1$

or, $2(\cot^2 A + \cot^2 B + \cot^2 C) = 2 \quad 1$

or, $2\cot^2 A + 2\cot^2 B + 2\cot^2 C - 2 \quad 1 = 0$

or, $2\cot^2 A + 2\cot^2 B + 2\cot^2 C$

$-2(\cot A \cot B + \cot B \cot C + \cot C \cot A) = 0$ [using (1)]

or, $\cot^2 A + \cot^2 B - 2\cot A \cot B + \cot^2 B + \cot^2 C - 2\cot B \cot C + \cot^2 C + \cot^2 A - 2\cot C \cot A = 0$

or, $(\cot A - \cot B)^2 + (\cot B - \cot C)^2 + (\cot C - \cot A)^2 = 0$

which is possible only when,

$(\cot A - \cot B)^2 = 0$ i.e. $\cot A = \cot B$

$(\cot B - \cot C)^2 = 0$ i.e. $\cot B = \cot C$

and $(\cot C - \cot A)^2 = 0$ i.e. $\cot C = \cot A$

Hence, $\cot A = \cot B = \cot C$

or, $A = B = C$

Hence, ΔABC is equilateral.

Illustration 85

If $A + B + C = \pi$, prove that

$$\cos A + \cos B + \cos C - 1 = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

Solution :

L.H.S. $= (\cos A + \cos B) + \cos C - 1$

$$= 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + \cos C - 1$$

$$= 2\cos\left(\frac{\pi}{2} - \frac{C}{2}\right)\cos\frac{A-B}{2} + \cos C - 1$$

$$= 2\sin\frac{C}{2}\cos\frac{A-B}{2} + 1 - 2\sin^2\frac{C}{2} - 1$$

$$= 2\sin\frac{C}{2}\cos\frac{A-B}{2} - 2\sin^2\frac{C}{2} = 2\sin\frac{C}{2}\left[\cos\frac{A-B}{2} - \sin\frac{C}{2}\right]$$

$$= 2\sin\frac{C}{2}\left[\cos\frac{A-B}{2} - \sin\left(\frac{\pi}{2} - \frac{A+B}{2}\right)\right] = 2\sin\frac{C}{2}\left[\cos\frac{A-B}{2} - \cos\frac{A+B}{2}\right]$$

$$= 2\sin\frac{C}{2} \cdot \sin\frac{A}{2}\sin\frac{B}{2} = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

Illustration 86

$$\frac{\sin 2A + \sin 2B + \sin 2C}{\cos A + \cos B + \cos C - 1} = 8\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

Solution :

$\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$

and $\cos A + \cos B + \cos C - 1 = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

dividing, we get $\dfrac{\sin 2A + \sin 2B + \sin 2C}{\cos A + \cos B + \cos C - 1}$

$$= \frac{4\sin A \sin B \sin C}{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= \frac{4.2\sin\frac{A}{2}\cos\frac{A}{2} \cdot 2\sin\frac{B}{2}\cos\frac{B}{2} \cdot 2\sin\frac{C}{2}\cos\frac{C}{2}}{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= 8\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$$

Illustration 87

If $A + B + C = \pi$, prove that $\cos\frac{A}{2} + \cos\frac{B}{2} + \cos\frac{C}{2} = 4\cos\frac{\pi - A}{4}\cos\frac{\pi - B}{4}\cos\frac{\pi - C}{4}$

Solution :

$$\text{L.H.S.} = \left(\cos\frac{A}{2} + \cos\frac{B}{2}\right) + \cos\frac{C}{2}$$

$$= 2\cos\frac{A+B}{4}\cos\frac{A-B}{4} + \cos\frac{C}{2}$$

$$= 2\cos\frac{\pi - C}{4}\cos\frac{A - B}{4} + \sin\left(\frac{\pi}{2} - \frac{C}{2}\right)$$

$$= 2\cos\frac{\pi - C}{4}\cos\frac{A - B}{4} + \sin\frac{\pi - C}{2}$$

$$= 2\cos\frac{\pi - C}{4}\cos\frac{A - B}{4} + 2\sin\frac{\pi - C}{4}\cos\frac{\pi - C}{4}$$

$$= 2\cos\frac{\pi - C}{4}\left[\cos\frac{A - B}{4} + \sin\frac{\pi - C}{4}\right]$$

$$= 2\cos\frac{\pi - C}{4}\left[\cos\frac{A - B}{4} + \cos\left(\frac{\pi}{2} - \frac{\pi - C}{4}\right)\right]$$

[on R. H. S. after leaving the factor $2\cos\frac{\pi - C}{4}$ we are left with $2\cos\frac{\pi - A}{4}\cos\frac{\pi - B}{4}$, therefore, we have changed sine in cosine. If both terms are in sine or cosine transform the terms in the bracket in the form cos C ± cosD, otherwise in the form sinC ± sinD]

$$= 2\cos\frac{\pi - C}{4}\left[\cos\frac{A - B}{4} + \cos\frac{\pi + C}{4}\right]$$

$$= 2\cos\frac{\pi - C}{4}2\cos\frac{\pi + A + C - B}{8}\cos\frac{\pi + C - A + B}{8}$$

$$= 4\cos\frac{\pi - C}{4}\cos\frac{\pi - B}{4}\cos\frac{\pi - A}{4} \qquad [\because A + C = +\pi - B \text{ and } B + C = \pi - A]$$

$$= 4\cos\frac{\pi - A}{4}\cos\frac{\pi - B}{4}\cos\frac{\pi - C}{4}$$

2nd Method :

$$\text{R.H.S.} = 4\cos\frac{\pi - A}{4}\cos\frac{\pi - B}{4}\cos\frac{\pi - C}{4}$$

$$= 2\cos\frac{\pi - A}{4}\left(2\cos\frac{\pi - B}{4}\cos\frac{\pi - C}{4}\right)$$

$$= 2\cos\frac{\pi - A}{4}\left[\cos\frac{2\pi - (B + C)}{4} + \cos\frac{C - B}{4}\right]$$

[Using the formula 2cosA cosB = cos(A + B) + cos(A – B)]

$$= 2\cos\frac{\pi - A}{4}\left[\cos\frac{2\pi - (\pi + A)}{4} + \cos\frac{C - B}{4}\right] \qquad [\because B + C = \pi - A]$$

$$= 2\cos\frac{\pi - A}{4}\left[\cos\frac{\pi + A}{4} + \cos\frac{C - B}{4}\right]$$

$$= 2\cos\frac{\pi - A}{4}\cos\frac{\pi + A}{4} + 2\cos\frac{\pi - A}{4}\cos\frac{C - B}{4}$$

$$= \cos\left(\frac{\pi - A}{4} + \frac{\pi + A}{4}\right) + \cos\left(\frac{\pi - A}{4} - \frac{\pi + A}{4}\right) + 2\cos\frac{C + B}{4}\cos\frac{C - B}{4} \quad [\because \pi - A = B + C]$$

$$= \cos\frac{\pi}{2} + \cos\left(-\frac{A}{2}\right) + \left[\cos\left(\frac{C + B}{4} + \frac{C - B}{4}\right) + \cos\left(\frac{C + B}{4} - \frac{C - B}{4}\right)\right]$$

$$= 0 + \cos\frac{A}{2} + \cos\frac{C}{2} + \cos\frac{B}{2} \quad [\because \cos(-\theta) = \cos\theta]$$

$$= \cos\frac{A}{2} + \cos\frac{B}{2} + \cos\frac{C}{2}$$

Illustration 88

In a triangle prove that

(a) cos A + cos B + cos C > 1 but not greater than 3/2

(b) $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \le \frac{1}{8}$

Solution :

(a) $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} > 1$ ...(1)

as neither of $\sin\frac{A}{2}, \sin\frac{B}{2}, \sin\frac{C}{2}$ is –ive or zero.

Again cosA + cosB + cosC

$$= 2\cos\frac{A + B}{2}\cos\frac{A - B}{2} + 1 - 2\sin^2\frac{C}{2} \le 2\sin\frac{C}{2}.1 + 1 - 2\sin^2\frac{C}{2}$$

$$\because \quad 0 \le \cos\frac{A - B}{2} \le 1$$

$$= -2\left[s^2 - s - \frac{1}{2}\right], \text{ where } s = \sin\frac{C}{2}$$

$$= -2\left[\left(s - \frac{1}{2}\right)^2 - \frac{1}{2} - \frac{1}{4}\right] = \frac{3}{2} - \left(s - \frac{1}{2}\right)^2 \le \frac{3}{2}$$

$\therefore \quad \cos A + \cos B + \cos C \leq 3/2$

In other words, $\sum \cos A \not> 3/2$.

(b) From above,

$$1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \leq \frac{3}{2}$$

$$\therefore \quad \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \leq \frac{1}{8}$$

Illustration 89

Prove that 1 + cos56 + cos58 – cos66 = 4cos28 cos29 sin33 **[I. I. T. 73]**

Solution :

L.H.S. $= 1 + \cos 56 + (\cos 58 - \text{os}66)$

$= 2\cos^2 28 + 2\sin 62 \sin 4$

$= 2\sin(90 - 28)\sin 4 + 2\cos^2 28$ $\quad [\because 62 = 90 - 28]$

$= 2\cos 28(\sin 4 + \cos 28)$

$= 2\cos 28(\sin 4 + \sin 62)$ $\quad [\because \cos 28 = \cos(90 - 62) = \sin 62]$

$= 2\cos 28 \cdot 2\sin 33 \cdot \cos 29$

$= 4\cos 28 \cdot \cos 29 \cdot \sin 33$

Illustration 90

If xy + yz + zx = 1, prove that

$$\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$$ **[I. I. T. 71]**

Solution :

Let $x = \tan A$, $y = \tan B$, $z = \tan C$

given $xy + yz + yz = 1$

$\therefore \quad \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$

or, $\quad \tan C(\tan A + \tan B) = 1 - \tan A \tan B$

or, $\quad \dfrac{\tan A + \tan B}{1 - \tan A \tan B} = \dfrac{1}{\tan C} = \cot C = \tan\left(\dfrac{\pi}{2} - C\right)$

or, $\quad \tan(A + B) = \tan\left(\dfrac{\pi}{2} - C\right)$

$\therefore \quad A + B = \frac{\pi}{2} - C$ [Taking principal value]

or, $A + B + C = \frac{\pi}{2}$ ...(1)

Now, L.H.S.

$$= \frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2}$$

$$= \frac{\tan A}{1-\tan^2 A} + \frac{\tan B}{1-\tan^2 B} + \frac{\tan C}{1-\tan^2 C}$$

$$= \frac{1}{2}\left[\frac{2\tan A}{1-\tan^2 A} + \frac{2\tan B}{1-\tan^2 B} + \frac{2\tan C}{1-\tan^2 C}\right]$$

$$= \frac{1}{2}(\tan 2A + \tan 2B + \tan 2C). \quad ...(2)$$

Now from (1), $A + B + C = \frac{\pi}{2}$

$\therefore \quad 2A + 2B + 2C = \pi$

or, $2A + 2B = \pi - 2C$ or, $\tan(2A + 2B) = \tan(\pi - 2C)$

or, $\frac{\tan 2A + \tan 2B}{1 - \tan 2A \tan 2B} = \tan 2C$

or, $\tan 2A + \tan 2B = -\tan 2C + \tan 2A \tan 2B \tan 2C$

or, $\tan 2A + \tan 2C + \tan 2B = \tan 2A \tan 2B \tan 2C$

From (2), L.H.S. $= \frac{1}{2}(\tan 2A + \tan 2B + \tan 2C)$

$$= \frac{1}{2}(\tan 2A \tan 2B \tan 2C)$$

$$= \frac{1}{2} \cdot \frac{2\tan A}{1-\tan^2 A} \cdot \frac{2\tan B}{1-\tan^2 B} \cdot \frac{2\tan C}{1-\tan^2 C}$$

$$= \frac{1}{2} \cdot \frac{2x}{1-x^2} \cdot \frac{2y}{1-y^2} \cdot \frac{2z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$$

Illustration 91

If $x + y + z = xyz$**, show that**

$$\frac{3x - x^3}{1 - 3x^2} + \frac{3y - y^3}{1 - 3y^2} + \frac{3z - z^3}{1 - 3z^2} = \frac{3x - x^3}{1 - 3x^2} \cdot \frac{3y - y^3}{1 - 3y^2} \cdot \frac{3z - z^3}{1 - 3z^2}$$

Solution :

Let $x = \tan A$, $y = \tan B$, $z = \tan C$

Now, $\therefore$ $x + y + z = xyz$

$\therefore$ $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

or, $\tan A + \tan B = -\tan C\,(1 - \tan A \tan B)$

or, $\dfrac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C = \tan(\pi - C)$

or, $\tan(A + B) = \tan(\pi - C)$

$\therefore$ $A + B = \pi - C$ [Taking principal value]

or, $A + B + C = \pi$ ...(1)

Now, L.H.S. $= \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A} + \dfrac{3\tan B - \tan^3 B}{1 - 2\tan^2 B} + \dfrac{3\tan C - \tan^3 C}{1 - 3\tan^2 C}$

$= \tan 3A + \tan 3B + \tan 3C$ ...(2)

Now, from (1), $A + B + C = \pi$ $\therefore$ $3A + 3B = 3\pi - 3C$

$\therefore$ $\tan(3A + 3B) = \tan(3\pi - 3C)$

$= \tan[2\pi + (\pi - 3C) = \tan(\pi - 3C)$

or, $\dfrac{\tan 3A + \tan 3B}{1 - \tan 3A \tan 3B} = -\tan 3C$

or, $\tan 3A + \tan 3B = -\tan 3C + \tan 3A \tan 3B \tan 3C$

or, $\tan 3A + \tan 3B + \tan 3C = \tan 3A \tan 3B \tan 3C$

$\therefore$ From (2) L.H.S. $= \tan 3A \tan 3B \tan 3C$

$$= \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A} \cdot \frac{3\tan B - \tan^3 B}{1 - 3\tan^2 B} \cdot \frac{3\tan C - \tan^3 C}{1 - 3\tan^2 C}$$

$$= \frac{3x - x^3}{1 - 3x^2}, \frac{3y - y^3}{1 - 3y^2} \cdot \frac{3z - z^3}{1 - 3z^2} = \text{R.H.S.}$$

Illustration 92

If A + B + C + D = 2π, prove that cosA + cosB + cosC + cosD = $4\cos\frac{A+B}{2}\cos\frac{B+C}{2}\cos\frac{C+A}{2}$

Solution :

L.H.S. $= (\cos A + \cos B) + (\cos C + \cos D)$

$$= 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + \cos\frac{C+D}{2}\cos\frac{C-D}{2}$$

$$= 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + \cos\left(\pi - \frac{A+B}{2}\right)\cos\frac{C-D}{2}$$

$$= \left[\because A+B+C+D = 2\pi \quad \therefore \frac{C+D}{2} = \pi - \frac{A+B}{2}\right]$$

$$= 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} - 2\cos\frac{A+B}{2}\cos\frac{C-D}{2}$$

$$= 2\cos\frac{A+B}{2}\left[\cos\frac{A-B}{2} - \cos\frac{C-D}{2}\right]$$

$$= 2\cos\frac{A+B}{2} \times 2\sin\frac{A-B+C-D}{4}\sin\frac{C-D-A+B}{4}$$

$$= 4\cos\frac{A+B}{2}\sin\frac{A+C-(B+D)}{4}\sin\frac{B+C-(A+D)}{4}$$

$$= 4\cos\frac{A+B}{2}\sin\frac{A+C-(2\pi-A-C)}{4}\sin\frac{B+C-(2\pi-B-C)}{4}$$

$[\because \quad A+B+C+D = 2\pi \quad \therefore \quad B+D = 2\pi - A - C \quad$ and $\quad A+D = 2\pi - B - C]$

$$= 4\cos\frac{A+B}{2}\sin\left(\frac{A+C}{2} - \frac{\pi}{2}\right)\sin\left(\frac{B+C}{2} - \frac{\pi}{2}\right)$$

$$= 4\cos\frac{A+B}{2}\left[-\sin\left(\frac{\pi}{2} - \frac{A+C}{2}\right)\right] \times \left[-\sin\left(\frac{\pi}{2} - \frac{B+C}{2}\right)\right] \qquad [\because \sin(-\theta) = -\sin\theta]$$

$$= 4\cos\frac{A+B}{2}\cos\frac{A+C}{2}\cos\frac{B+C}{2}$$

Illustration 93

If A + B + C = 2S, prove that

$$\cos^2 S + \cos^2(S - A) + \cos^2(S - B) + \cos^2(S - C) = 2 + 2\cos A \cos B \cos C$$

Solution :

$$\text{L.H.S.} = \cos^2 S + \cos^2(S - A) + \cos^2(S - B) + \cos^2(S - C)$$

$$= \frac{1 + \cos 2S}{2} + \frac{1 + \cos 2(S - A)}{2} + \frac{1 + \cos 2(S - B)}{2} + \frac{1 + \cos 2(S - C)}{2}$$

$$= \frac{1}{2}[4 + \{\cos 2S + \cos(2S - 2A\} + \{\cos(2S - 2B) + \cos(2S - 2C)]\}$$

$$= \frac{1}{2}\left(4 + 2\cos\frac{4S - 2A}{2}\cos\frac{2A}{2} + 2\cos\frac{4S - 2B - 2C}{2}\cos\frac{2(C - B)}{2}\right)$$

$$= \frac{1}{2}[4 + 2\cos(2S - A)\cos A + 2\cos(2S - B - C)\cos(C - B)]$$

$$= \frac{1}{2}[4 + 2\cos(B + C)\cos A + 2\cos A\cos(C - B)]$$

$[\therefore\ A + B + C = 2S \qquad \therefore \qquad 2S - A = B + C \text{ and } 2S - B - C = A]$

$$= \frac{1}{2}[4 + 2\cos A\{\cos(B + C) + \cos(C - B)]\}$$

$$= \frac{1}{2}[4 + 2\cos A \times 2\cos B\cos C]$$

$$= 2 + 2\cos A \cos B \cos C$$

Illustration 94

If A + B + C = π, show that $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$

Solution :

$\because$ A + B + C = π then as done earlier

$$\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1 \qquad ...(1)$$

Let $\tan\frac{A}{2} = x, \tan\frac{B}{2} = y, \tan\frac{C}{2} = z$

then from (1), $xy + yz + zx = 1$ ...(2)

Now $\because$ $(x - y)^2 + (y - z)^2 + (z - x)^2 \geq 0$

$\because$ $x^2 + y^2 - 2xy + y^2 + z^2 - 2yz + z^2 + x^2 - 2zx \geq 0$

or, $2(x^2 + y^2 + z^2) \geq 2(xy + yz + zx)$

or, $x^2 + y^2 + z^2 \geq 1$ $[\because xy + yz + zx = 1)$

Putting the values of x, y, z.

We get $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$

Illustration 95

If A + B + C = π, prove that

(tanA + tanB + tanC) (cotA + cotB + cotC) = 1 + secA secB secC

Solution :

$\therefore$ $A + B + C = \pi$ $\therefore$ $A + B = \pi - C$

$\tan(A + B) = \tan(\pi - C) = -\tan C$

or $\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$. which on simplification gives

$\tan A + \tan B + \tan C = \tan A \tan B \tan C$ ...(1)

Now, L.H.S. $= \tan A \tan B \tan C (\cot A + \cot B + \cot C)$

$= \tan B \tan C + \tan C \tan A + \tan A \tan B$

$[\because \tan A \cot A = 1, \tan B \cot B = 1, \tan C \cot C = 1]$

$$= \frac{\sin B \sin C}{\cos B \cos C} + \frac{\sin C \sin A}{\cos C \cos A} + \frac{\sin A \sin B}{\cos A \cos B}$$

$$= \frac{\cos A \sin B \sin C + \cos B \sin C \sin A + \cos C \sin A \sin B}{\cos A \cos B \cos C}$$

$$= \frac{\sin C[\cos A \sin B + \sin A \cos B] + \cos C \sin A \sin B}{\cos A \cos B \cos C}$$

$$= \frac{\sin C \sin(A + B) + \cos C \sin A \sin B}{\cos A \cos B \cos C} = \frac{\sin C \sin(\pi - C) + \cos C \sin A \sin B}{\cos A \cos B \cos C}$$

$$= \frac{\sin^2 C + \cos C \sin A \sin B}{\cos A \cos B \cos C} \qquad [\therefore \sin(\pi - C) = \sin C]$$

$$= \frac{1 - \cos^2 C + \cos C \sin A \sin A}{\cos A \cos B \cos C} = \frac{1 + \cos C(\sin A \sin B - \cos C)}{\cos A \cos B \cos C}$$

$$= \frac{1 + \cos C[\sin A \sin B - \cos\{\pi - (A + B)\}]}{\cos A \cos B \cos C}$$

$$= \frac{1 + \cos C[\sin A \sin B + \cos(A + B)]}{\cos A \cos B \cos C}$$

$$= \frac{1 + \cos C[\sin A \sin B + \cos A \cos B - \sin A \sin B]}{\cos A \cos B \cos C}$$

$$= \frac{1 + \cos C \cos A \cos B}{\cos A \cos B \cos C} = \sec A \sec B \sec C + 1$$

Illustration 96

If A + B + C = π, prove that

(cotB + cotC) (cotC + cotA) (cotA + cotB) = cosecA cosecB cosecC

Solution :

$$\cot B + \cot C = \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}$$

$$= \frac{\cos B \sin C + \sin B \cos C}{\sin B \sin C} = \frac{\sin(B + C)}{\sin B \sin C}$$

$$= \frac{\sin(\pi - A)}{\sin B \sin C} = \frac{\sin A}{\sin B \sin C} \quad ...(1)$$

Similarly, $\cot C + \cot B = \dfrac{\sin C}{\sin A \sin B}$

and $\cot A + \cot B = \dfrac{\sin C}{\sin A \sin B}$

Multiplying (1), (2) and (3), we get

$$\text{L.H.S.} = \frac{\sin A}{\sin B \sin C} \cdot \frac{\sin B}{\sin C \sin A} \cdot \frac{\sin C}{\sin A \sin B}$$

$$= \text{cosecA cosecB cosecC}$$

Illustration 97

If A + B + C = π, show that $\frac{1}{2}\sum \sin^2 A(\sin 2B + \sin 2C) = 2\sin A \sin B \sin C$

Solution :

Ist term = $\sin^2 A \sin B \cos B + \sin^2 A \sin C \cos C$

Similarly, second term = $\sin^2 B \sin C \cos C + \sin^2 B \sin A \cos A$

and third term = $\sin^2 C \sin A \cos A + \sin^2 C \sin B \cos B$

Now, L.H.S. = $(\sin^2 A \sin B \cos B + \sin^2 B \sin A \cos A) + (\sin^2 A \sin C \cos C + \sin^2 C \sin A \cos A)$
$+ (\sin^2 B \sin C \cos C + \sin^2 C \sin B \cos B)$

$= \sin A \sin B \sin(A + B) + \sin A \sin C \sin(A + C) + \sin B \sin C (B + C)$

$= \sin A \sin B \sin C + \sin A \sin B \sin C + \sin A \sin B \sin C$

$= 3\sin A \sin B \sin C.$